

BANK OF THAILAND

Monetary Policy Report

March 2015

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2015

Mr. Prasarn Trairatvorakul	Chairman
Mrs. Pongpen Ruengvirayudh	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Veerathai Santiprabhob	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim of ensuring that monetary policy decisions are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

On January 6, 2015 the Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the monetary policy target for 2015, in place of the quarterly average core inflation target of 0.5 – 3.0 percent. The new inflation target was jointly proposed by the MPC and the Minister of Finance. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the public, together with measures taken and estimated time to bring inflation back to the target.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy expanded by slightly less than the previous projection due to weaker-than-anticipated domestic demand momentum in 2014 Q4, along with weakening global growth and less-than-expected public spending. Looking ahead, lower global oil prices will likely support the recovery of private spending. Growth is projected to pick up pace in 2016, on a lower drag from household debt, greater clarity on public spending, and an improvement in exports. Meanwhile, headline inflation in 2015 is revised down significantly on the back of lower oil prices, but is likely to rebound in the second half of 2015 and in 2016, in line with oil price outlook.

Against the backdrop of weaker-than-expected economic recovery and higher downside risks to growth, additional support from monetary policy is necessary. Meanwhile, despite the forecast that headline inflation could breach the lower bound of the target band, the Monetary Policy Committee considers this a result of falling energy prices, not a case of deflation caused by aggregate demand contraction.

1.1 Growth and inflation prospects

The Committee revised down Thailand's economic growth forecast in 2015 from 4.0 to 3.8 percent, in the light of the lower-than-projected private spending in late 2014. Weak private spending was caused by waning private sector confidence, amidst concerns over uncertainties of future income. Businesses also delayed their investment as they awaited a clearer sign of economic recovery. Public spending is projected to edge up, but limitations remain in budget disbursements. Exports are assessed to grow slightly below the previous projection, with export volumes declining in line with slowing

global growth. Export prices, especially commodity prices, are also projected to fall due to feeble global demand and lower global oil prices. However, the prospect of solid tourism growth will provide a boost to the economy.

Table 1.1 Forecast Summary

Percent	2014*	2015	2016
GDP growth	0.7	3.8	3.9
		(4.0)	
Headline inflation	1.9	0.2	2.2
		(1.2)	
Core inflation	1.6	1.2	1.2
		(1.2)	

Note: * Outturn
( ) MPR Dec 2014
Source: Office of the National Economic and Social Development Board and calculations by Bank of Thailand

The Thai economy in 2016 is projected to expand by 3.9 percent, due to a waning drag from household debt and a clearer prospect of public expenditure reforms designed to promote long-term investment. Exports are likely to resume its role as a driver of growth, while a rebound in domestic and external demand should help shore up private sector confidence and boost private investment going forward **(Tables 1.1 and 1.4)**.

Headline inflation forecast in 2015 is adjusted down from 1.2 to 0.2 percent, below the lower bound of the new inflation target band of 2.5 $\pm$ 1.5 percent. This adjustment follows substantial falls in global oil prices in recent periods. Nevertheless, in the MPC's view, the latest inflation forecast does not signify deflation because the prices of most goods continued to rise or remained stable. Housing rents also increased in early 2015, pushing up core inflation to 1.2 percent. Public expectation of medium-term inflation remained well anchored close to the central target of headline inflation. These reasons were specified earlier in an Open Letter issued to the public following the negative headline inflation in January 2015 **(Article: "Inflation Target in 2015")**. Moreover, headline inflation is likely to remain low and negative in the first half of the year, before rebounding in the latter half based on the Committee's projection of oil prices.

The headline inflation forecast for 2016 is 2.2 percent, based on the assumptions that oil and fresh food prices will increase from their 2015

levels while the core inflation forecast for 2016 is 1.2 percent. Although inflationary pressure from the demand side is likely to be lower than previously projected, it is still forecasted to increase from last year. This is reflected by the gradual narrowing of the output gap in line with economic recovery **(Chart 1.1)**.

The Committee factored the following observations into the growth and inflation forecasts for 2015 and 2016.

(1) Global economic recovery is projected to be slower than previously forecast, due to a slowdown in the Chinese and Asian economies (Table 1.2).

Sluggish global recovery, led by a slowdown of the Chinese economy, prompted a downward revision to the Committee's forecast of merchandise export growth. Meanwhile, supply-side constraints in the Thai manufacturing sector are likely to weigh on merchandise exports, with Thai manufacturers still producing goods that do not respond to the changing global demand. Exports of services, however, are projected to post a strong growth. In addition, despite the economic slowdown in China, demand for foreign travels among Chinese tourists is assessed to remain strong. This should therefore help compensate for the adverse impacts caused by fewer visitors from Russia, Europe and oil-exporting countries.

Moreover, the prices of Thai agricultural products are likely to be held down by moderating external demand. As China is a major importer of commodities including agricultural commodities, weakening Chinese growth will have a negative impact on these prices. Depressed agricultural prices will in turn hit farm incomes and undermine household confidence. Private consumption is thus

Chart 1.1 Output Gap

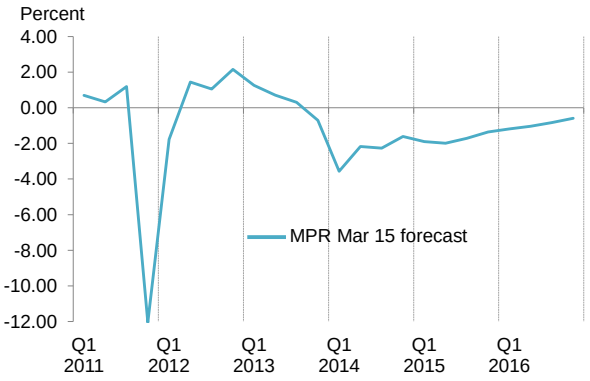


Table 1.2 Growth Assumptions for Thailand's Trading Partners

Annual percentage change (%YoY)	Weight (%)	2014	2015		2016
			Dec 2014	Mar 2015	Mar 2015
The U.S.	14.3	2.4	3.1	3.2	2.9
The euro area	10.3	0.9	0.9	1.1	1.5
Japan	14.4	0.0	1.0	1.0	1.4
China	15.2	7.4	7.2	7.0	6.9
Asia (excluding Japan and China)*	36.6	4.1	4.4	4.2	4.4
Total*	100	3.5	3.8	3.7	3.9

Note: * Weighted by each trading partner's share in Thailand's total exports in 2010 (7 countries: Singapore (6.4%), Hong Kong (9.3%), Malaysia (7.5%), Taiwan (2.3%), Indonesia (5.2%), South Korea (2.6%) and the Philippines (3.5%)

** Weighted by each trading partner's share in Thailand's total exports in 2010 (13 countries)

likely to expand at a slower rate than previously assessed.

(2) Domestic demand momentum in 2014 Q4 was weaker than expected.

Private spending expanded at a lower rate than the previous assessment, reflecting fragile private sector confidence amidst concerns over weak economic recovery. Subdued farm incomes and elevated household debt caused households to be cautious with their spending, especially on durable goods. Several industries still had excess production capacity amidst sluggish recovery of both domestic and external demand, leaving little justification for new private investment. This was consistent with the business sentiment survey, which suggested most firms chose to postpone their investment while awaiting greater clarity on the economic recovery and public investment in infrastructure.

Moreover, financial institutions remained cautious in lending to the private sector, especially households and small and medium enterprises (SMEs). Although monetary policy continued to be accommodative, the strict lending by banks restrained household and business purchasing power, contributing to the slow recovery of domestic spending.

The Committee thus revised down domestic demand momentum in the forecast period, while expressing concerns over muted business confidence which, if prolonged, could further discourage productivity-enhancing investment and undermine Thailand's long-term potential and competitiveness.

(3) Less-than-expected public spending, particularly public investment.

The Committee lowered the forecast of the government's budget disbursement rate for fiscal year 2015 from 93 to 91.2 percent, largely due to sluggish public investment. In the budget setting process, the government planned to implement public spending reforms aimed at increasing investment expenditure to support infrastructure development. However, fiscal stimulus measures have encountered short-term delays due to (1) limitations in the budget disbursement process, especially in investment expenditure and (2) the introduction of lower construction costs used for government procurement following lower oil prices. Disbursement rate for the fiscal year 2016 are forecasted at 90.5 percent because (1) an improvement in budget disbursement might not be able to keep pace with the simultaneous increase in public investment budgets and (2) there is a continued shortage of labor in the construction sector **(Table 1.3)**.

Table 1.3 Assumptions on public sector expenditure

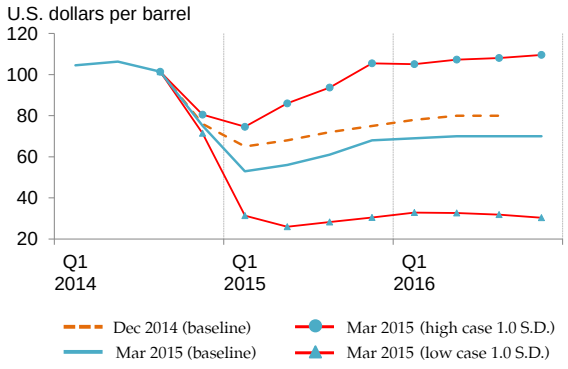
Unit: Billion baht	Fiscal year	
	2015	2016
General government consumption	1,802.0	1,874.3
Public investment	683.9	734.0
Total	2,485.9	2,608.3

Note: Includes expenditure assumptions on the water management project.
Source: Forecast by Bank of Thailand

(4) The decline in global oil prices during Q4 2014 and early 2015.

The Committee revised down the baseline assumption for global oil price to 60 U.S. dollars per barrel in 2015 and 70 U.S. dollars per barrel in 2016 **(Chart 1.2)**, on the back of substantial decline in oil prices in recent periods. Nevertheless, oil prices are projected to gradually increase in the latter half of 2015. The Committee judged that falling global oil prices will have the following impacts: (1) Lower domestic retail oil prices caused headline inflation in 2015 to move significantly lower than previously forecast; (2) Private spending should pick up on the back of lower costs of living and production costs, thereby increasing purchasing power and will be reflected by growing private consumption in the periods ahead. (3) Low global oil prices will put downward

Chart 1.2 Assumptions on Dubai Oil Price

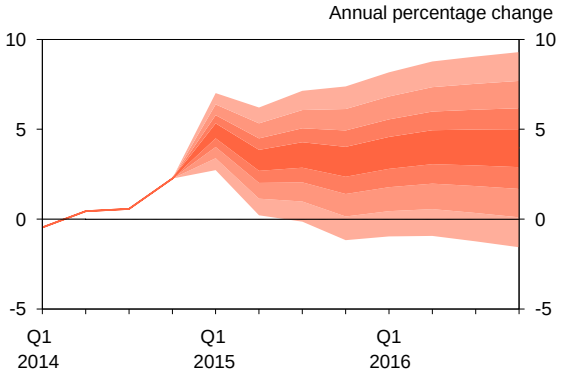


pressure on the prices of other commodities that move together with oil prices. Therefore, the prices of these commodities, particularly petroleum, chemicals and rubber, will likely remain low and slightly below the previous projection. Given that these commodities account for 18.4 percent of total Thai exports, the growth of merchandise exports is revised down from the previous assessment. (4) As a net oil importer, Thailand benefits from lower oil prices through markedly lower oil import values. **This, in turn, should translate into a large current account surplus in 2015 (Table 1.4), while contributing to the baht's strength against regional currencies in recent periods.** Looking ahead, current account is expected to move closer to the equilibrium in 2016. Imports are projected to increase on the back of a rebound in domestic spending and assumption of higher global oil prices compared to the 2015 level **(Table 1.5)**.

Downside Risks to Growth and Inflation Forecasts

According to the MPC's assessment, **the probability that growth will be below baseline projection is higher than the probability that it will be above the baseline projection. This assessment is shown in the growth fan chart, which is skewed downward throughout the forecast period (Chart 1.3 and Table 1.6).**

Chart 1.3 GDP Growth Forecast



Note: The fan chart covers 90 percent of the probability distribution.

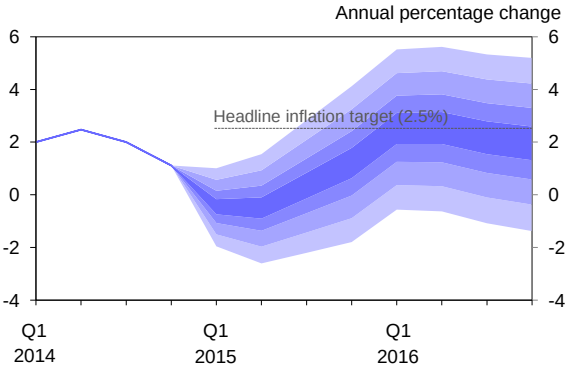
Downside risks that could lead economic growth to be lower than the baseline projection stem from the following: (1) The pace of global economic recovery could be slower than anticipated. Economic conditions in the euro area remain fragile, plagued by the ongoing political uncertainty in Greece and the geopolitical conflict concerning Russia. This factor, coupled with the slowdown in

the Chinese economy, could weigh on Thai exports and continue to depress agricultural prices and farm incomes. Meanwhile, private sector confidence remains weak amidst sluggish economic recovery. Therefore, private spending could turn out to be lower than baseline projection. (2) Public spending could fall short of expectation because disbursements might not be able to keep pace with budget expansion, especially with respect to public infrastructure investment projects.

Risk factors that could cause the economy to expand at a rate higher than the baseline projection could arise from the following sources: (1) Public spending could exceed the previous forecast due to the second-round fiscal stimulus measures, including the water management system project and the urgent road transport infrastructure development plan. (2) Domestic retail oil prices could fall below the baseline assumption and therefore provide further boost to household spending.

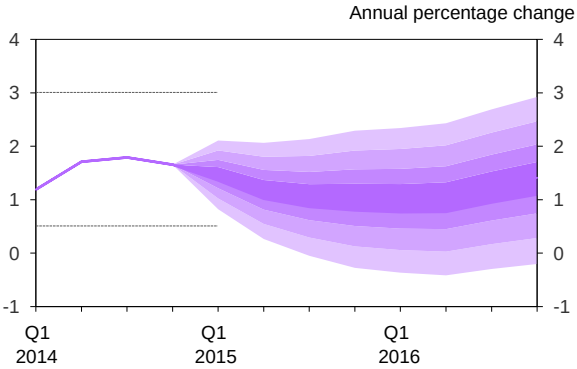
In the light of these risk factors, the Committee judges that **headline and core inflation are more likely to fall below the central projection than to surpass it. This is reflected in the inflation fan charts that are skewed downward throughout the forecast period (Charts 1.4 and 1.5, Tables 1.7 and 1.8).** The assessment stems from the possibility that the government may cut the diesel and gasohol contribution rates to the Oil Fund, in the light of the fund's stronger balance. In addition, some Committee members are of the view that global oil prices could fall below the current assumptions, while domestic demand momentum may be more subdued than the projection if economic growth turns out to be weaker than expected.

Chart 1.4 Headline Inflation Forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.5 Core Inflation Forecast



Note: The fan chart covers 90 percent of the probability distribution.

1.2 Monetary policy decision

Monetary policy has become more accommodative.

Monetary policy has played a greater role in supporting the Thai economy during 2015 Q1, amidst weaker-than-expected economic recovery and higher downside risks to growth. While headline inflation is expected to breach the lower bound of the target band, the Committee considers this a result of positive supply-side shocks associated with lower energy prices. Therefore, the lower inflation forecast is not deemed a sign of deflation stemming from aggregate demand contraction.

At its meeting on January 28, 2015, the MPC voted 5 to 2 to maintain the policy interest rate at 2.00 percent, with two members in favor of a reduction of the policy rate by 0.25 percent. The Committee deliberated on the importance of public spending in driving the overall economic recovery. Moreover, the Committee also discussed the implications that less accommodative monetary conditions and the possibility of inflation breaching the target band would have on monetary policy conduct. The Committee agreed that clear and consistent public spending plans, especially investment projects, would be highly effective in driving the growth momentum.

The majority of the Committee members deemed the level of policy rate appropriate in maintaining the consistency of monetary policy, given that economic conditions had remained largely unchanged from the previous meeting. Despite the slow pace of growth, the economy remained firmly on the path of steady recovery.

Meanwhile, substantially lower oil prices would lend support to the rebound of domestic demand. Downside risks to the economy were generally well contained. Given this economic outlook, the 2.00 percent policy rate did not hinder the ongoing economic recovery. Real interest rate, calculated based on inflation forecasts, remained low. Besides, lower borrowing costs in the bond markets should further boost private sector financing. Although the easing of monetary policy by foreign central banks could encourage capital inflows to the Thai financial markets and put upward pressure on the baht, no significant inflows had been observed to date. This was partly due to the low yields on the Thai government bonds relative to those in other regional countries. Moreover, the Committee viewed the possible breach of the headline inflation target to be caused by supply-side shocks from lower energy prices, which in turn should contribute to stronger economic recovery in the periods ahead. The lower inflation forecast was not an indication of deflation arising from a contraction of aggregate demand, as confirmed by the fact that core inflation remained positive and quite stable. In addition, public expectation of medium-term inflation was well anchored around the central target. This indicated that the public shared the MPC's assessment that the negative headline inflation reading was temporary in nature. Under these circumstances and with no deflation risks, additional easing of monetary policy was therefore considered unnecessary.

Nevertheless, the minority of the Committee deemed it necessary to further ease monetary policy. Their views were influenced by the substantially and persistently lower-than-potential growth performance of the Thai economy and limitation of fiscal stimulus. Global recovery was

also projected to slow with greater downside risks. Moreover, a policy rate cut might help cushion the impacts of easy monetary conditions abroad and reduce some of the pressure on the baht. This could stem the baht appreciation against the currencies of Thailand's main trading partners and competitors. Given the recent move to headline inflation targeting, reducing the policy rate should help bolster the credibility of monetary policy framework, at the time when headline inflation moved below the lower bound of the target band.

Subsequently at the meeting on March 11, 2015, the MPC voted 4 to 3 to lower the policy interest rate by 0.25 percent, from 2.00 percent to 1.75 percent, with three members voting to keep the rate on hold. The Committee's deliberation focused on the appropriate role of accommodative monetary policy, given that the Thai economy was projected to recover at a slower pace than previously assessed. The Committee also considered the limitation of fiscal stimulus, the subdued inflation forecasts in the periods ahead, and the effectiveness and potential costs of additional easing of monetary policy under the current circumstances.

The majority of the Committee members judged that monetary policy should be further loosened to provide more support to economic growth and shore up private sector confidence, in the light of the weakening momentum from private consumption and investment. The reduction in the policy rate would help ease monetary conditions. Meanwhile, risks to financial stability remained contained, as seen by the recent decline in price-earnings ratio (P/E ratio) following new regulatory measures and private sector's waning debt accumulation in line with soft economic conditions.

Nonetheless, the minority of the Committee members were of the view that the current policy rate was sufficiently accommodative for bolstering economic recovery. In their views, the current policy rate was low relative to that of regional countries, and did not hinder economic activities. Moreover, they believed that monetary policy space should be preserved for future use, should more needs arise and when policy transmission becomes more effective. Fiscal stimulus, especially the implementation of planned public investment, should be a key growth driver at this juncture. In addition, further easing of monetary policy might lead to more financial imbalances such as a higher level of household debt, undermining long-term financial stability.

Going forward, the Committee will closely monitor the progress of economic recovery and stand ready to take appropriate policy actions to support a steady recovery of the Thai economy, while ensuring long-term financial stability.

1.3 Appendix

Table 1.4 Forecasts for GDP and Components			
Percent (per annum)	2014*	2015	2016
GDP growth	0.7	3.8	3.9
Domestic demand	-0.2	3.1	4.5
Private consumption	0.3	2.4	3.8
Private investment	-1.9	3.1	8.0
Government consumption	2.8	4.2	2.1
Public investment	-6.1	8.0	6.1
Exports of goods and services	0.0	3.6	5.4
Imports of goods and services	-4.8	4.4	6.8
Current Account (billion US dollars)**	14.2	16.5	8.5
Value of merchandise exports**	-0.3	0.8	4.0
Value of merchandise imports**	-8.5	0.0	8.8

Note: *Outturn
**Based on BPM6 definition

Table 1.5 Forecast Assumptions			
Annual percentage change	2014	2015	2016
Dubai oil price (U.S. dollars per barrel)	96.8	60	70
Non-fuel commodity prices (%YoY)	-3.9	-6.9	2.0
Fresh food prices (%YoY)	4.8	-4.2	2.8
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (%YoY)[1/]	5.3	5.3	3.8
Public investment (%YoY)[1/]	-4.1	9.9	7.5
Fed Funds rate (% at year-end)	0.13	0.88	2.38
Trading partners' economic growth (%YoY)[2/]	3.5	3.7	3.9
Regional currencies vis-à-vis the U.S. dollar (Index)[3/]	133.5	143.1	142.1

Note: [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by the minus sign

Table 1.6 Probability distribution of GDP growth forecast								
Percent	2015				2016			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 12	0	0	0	0	0	0	0	1
10-12	0	0	0	0	1	2	2	2
8-10	1	0	2	3	5	6	7	7
6-8	19	6	11	10	14	16	16	16
4-6	56	27	29	24	25	25	24	23
2-4	24	41	34	30	27	25	24	23
0-2	1	21	19	22	18	16	16	16
(-2)-0	0	4	5	9	8	7	8	8
< (-2)	0	0	1	3	2	2	3	4

Table 1.7 Probability distribution of headline inflation forecast								
Percent	2015				2016			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 7	0	0	0	0	1	1	1	0
6-7	0	0	0	0	2	2	2	1
5-6	0	0	0	1	6	6	4	4
4-5	0	0	1	4	12	12	10	9
3-4	0	0	3	9	18	18	16	15
2-3	0	2	9	17	21	21	20	19
1-2	4	9	19	22	18	18	19	19
0-1	25	22	25	20	12	12	14	15
(-1)-0	42	31	22	14	6	6	9	10
(-2) - (-1)	24	23	13	8	2	3	4	5
(-3) - (-2)	5	10	5	3	1	1	1	2
< (-3)	0	2	1	1	0	0	0	1

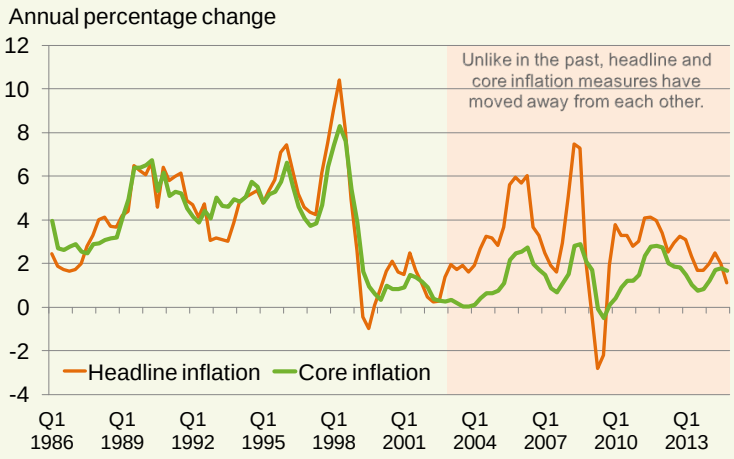
Table 1.8 Probability distribution of core inflation forecast								
Percent	2015				2016			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
>4.0	0	0	0	0	0	0	1	1
3.0-3.5	0	0	0	0	1	1	2	3
2.5-3.0	0	1	1	2	3	3	5	7
2.0-2.5	8	6	6	7	8	8	12	14
1.5-2.0	39	21	18	17	16	16	18	19
1.0-1.5	41	35	28	24	23	22	22	20
0.5-1.0	11	26	26	23	23	21	19	17
0.0-0.5	1	10	15	16	16	15	12	10
(-0.5)-0.0	0	2	5	7	8	8	6	5
(-1.0)-(-0.5)	0	0	1	2	3	3	2	2
< (-1.0)	0	0	0	1	1	1	1	1

Inflation target in 2015

An appropriate inflation target is key to the central bank's successful maintenance of price stability because it helps anchor public inflation expectations. It is thus essential for the central bank to set an inflation target that is consistent with the public understanding.

To that end, on December 17, 2014, the Monetary Policy Committee (MPC) proposed the annual average headline inflation target of 2.5 ± 1.5 percent[1/] as the monetary policy target for 2015, in place of the quarterly average core inflation target of 0.5 – 3.0 percent. The new monetary policy target was approved by the Cabinet on January 6, 2015. **The shift to headline inflation target should lead to improved monetary policy communication with the public and greater effectiveness in anchoring inflation expectations. Moreover, headline inflation is a better indicator of changes in the price level and the cost of living than core inflation**, as it covers all categories of goods and services consumed by the public, including energy and fresh food prices which account for a sizable 27 percent of the consumer basket. Energy and fresh food prices have had a significant influence on inflation in recent periods, but excluded from core inflation calculation. Research also shows that over the past ten years core inflation and headline inflation have been moving away from each other **(Chart 1)**.

Chart 1 Headline and core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Furthermore, **a mid-point target of 2.5 percent was introduced in place of an inflation target range of 0.5 – 3.0 percent.** This change will lead to clearer signaling of monetary policy to maintain price stability and improve the anchoring of inflation expectation. There is also a **tolerance band of ± 1.5 percent**, allowing flexibility for monetary policy implementation in order to meet the objectives of both output and price stability. At present, the tolerance band of 1.5

[1/] The MPC agrees to lower the mid-point target from 3.0 percent (MPC's decision on September 17, 2014) to 2.5 percent. This new target is considered a better reflection of substantially lower global inflation outlook that stemmed from structural changes in the oil market. Technological advancement in shale oil production in the U.S., coupled with the expansion of oil production in non-OPEC countries, result in greater ability of oil supply to readily meet oil demand. At the same time, OPEC countries suffer from declining market power. As a result, it is less likely that crude oil prices will sharply increase, as was the case in the 2000-2011 period.

percent is considered appropriate in the light of the volatility in energy and fresh food prices. **Time horizon is also extended,** with annual average inflation instead of the quarter average, in order to be consistent with the transmission lags of monetary policy.

In practice, the approach, framework and the decision-making process of monetary policy remain the same. The new inflation target still retains the essence of the previous regime. That is, the mid-point of the new headline inflation target (2.5 percent) is simply the sum of (1) the product of core inflation weight and the mid-point of the old inflation target range (0.73*1.75=1.28) and (2) the product of energy and fresh food prices weight and energy and food prices inflation (0.27*5.13=1.39). The sum is about 2.67 **((3)+(4) in Table 1)**. The policy deliberation process remains unchanged, with the MPC still attaching great importance to supporting sustained economic growth in line with its potential, without undermining price stability. The MPC also seeks to anchor public inflation expectations to an appropriate level, taking a forward-looking approach in responding to demand-pull inflationary pressures and inflation forecasts. In the process, headline inflation, core inflation, and other inflation indicators are also taken into consideration.

Table 1 Calculation of the mid-point of the new inflation target

	Weight in headline inflation basket	%YoY	Contribution (%)
	(1)	(2)	(1) x (2)
Core inflation (3)	0.73	1.75 (mid-point of the old target)	1.28
Energy and fresh food prices (4)	0.27	5.13*	1.39
Headline inflation (3) + (4)	1.00		2.67

Note: *Monthly average data in the pre-inflation targeting period (January 1989 – April 2000) was used because the period covered commodity price cycles.
Source: Ministry of Commerce, calculations by the Bank of Thailand

Nonetheless, the adoption of the new inflation target faces many challenges because **headline inflation is much more volatile in nature than core inflation**. Hence, headline inflation may sometimes breach the target as a result of changes in supply conditions beyond the control of monetary policy, especially changes in energy and fresh food prices. Monetary policy response may not be justified in the short term, if inflation indicators do not suggest worrying demand-pull inflationary pressures and if inflation forecasts remain appropriate. Nevertheless, **the MPC recognizes the importance of communicating such development to the public and the Ministry of Finance through an open letter.** In essence, the MPC clearly explained the reasons for the breach of inflation target, as well as the steps taken and time needed to bring inflation back to the target. In line with the practice of other central banks, the BOT's communication with the public in this manner should help anchor public inflation expectations.

Since 2014 Q4, global oil prices have substantially dropped. The resulting fall in domestic oil prices is the chief reason behind the moderation of headline inflation, which turned negative and breached the lower bound of the inflation target band in January 2015. Although the annual average inflation is not breached, the MPC wished to reaffirm its commitment to maintaining the monetary policy framework and anchoring public inflation expectations. Hence, to foster public understanding of the inflation situation, **the MPC submitted an open letter to the Ministry of Finance on February 2, 2015**, and published the letter on the BOT website.[2/] The key messages of the letter were as follows. First, headline inflation in January 2015 was below the lower bound of the target band because of lower global oil prices. Headline inflation is expected to remain negative until 2015 Q2, after which it was forecast to pick up and move within the band of the target by 2015 Q4. Second, the negative headline inflation was not an indication of deflation. Lower oil prices were also expected to lend support to economic recovery. Third, the MPC considered the decline in headline inflation a result of supply factors rather than sluggish demand. The present monetary policy stance was deemed adequately supportive of economic recovery and should help bring headline inflation back to within the target band by the end of this year.

[2/] The open letter is available on the Bank of Thailand website: https://www.bot.or.th/Thai/MonetaryPolicy/MonetPolicyKnowledge/AnnounceMPC/2558.pdf

Recent Economic Developments

2. Recent Economic Developments

The global economy recovered at a lower pace than previously assessed, owing to a slowdown in the Chinese and other Asian economies. The U.S. grew at a solid pace, while the euro area and the Japanese economies showed improvement from the previous quarter. Going forward, the plunge in global oil prices is expected to support global economic recovery.

In 2014 Q4, the Thai economy recovered at a gradual pace, held down by weak domestic spending associated with concerns over future income. Private investment also stalled amidst uncertainties over economic recovery and public investment, which was delayed by limitations in policy implementation. By contrast, exports of goods and services expanded more than expected, leading to an improvement in the manufacturing sector, while government spending edged up.

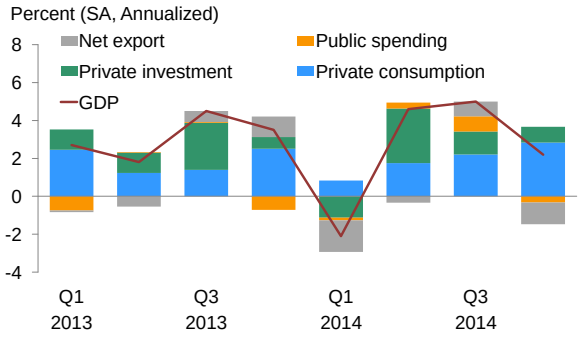
2.1 The global economy

The global economy expanded at a slower pace than assessed in the last monetary policy report due to slower growth in the Chinese and Asian economies. Meanwhile, the U.S. economy continued to post solid growth, while the euro area and the Japanese economies showed improvement from the previous quarter.

The U.S. economy exhibited robust growth stemming from continued improvement in economic fundamentals.

The U.S. Economy in 2014 Q4 grew by 2.2 percent (qoq saar, second estimate). The pace of recovery moderated from the previous quarter due to a decline in defense spending from three months earlier. Nonetheless, private consumption and investment showed signs of a steady recovery **(Chart 2.1).**

Chart 2.1 Contributions to U.S. Growth
(Change compared to the previous quarter)



Source: Bureau of Economic Analysis

In 2015 Q1, the U.S. economy is expected to continue expanding, given gradually improving economic fundamentals. Although a number of key economic indicators in January 2015, including Retail Sales, Manufacturing and Service PMI, suggested a decline in economic activity, they partly reflect corrections from steep accelerations of the previous period.

The U.S. monetary policy remained accommodative as the economy continued to expand, reflected by a faster-than-expected decline in unemployment rate. However, a number of labor market indicators indicated conflicting signals about the strength of the labor market recovery. For example, wages have not yet displayed significant improvement. Moreover, inflation reading was well below target as a result of the fall in global oil prices. The FOMC at its January 27 – 28, 2015 meeting maintained the Federal Funds Target rate at 0 – 0.25 percent.

The euro area economy[1/] recovered gradually, supported by lower oil prices and weakening euro. Business confidence also edged up, likely indicating continued growth in private consumption. Further risks remain, however, from the ongoing disagreements regarding the assistance plan for Greece.

The euro area economy in 2014 Q4 grew by 0.3 percent (qoq sa), up from 0.2 percent

[1/] The euro area economy consists of 18 countries that share the euro as an official currency. As of 2013, Germany, France, Italy, and Spain contributed to 28, 21, 17, and 11 percent of the economy respectively, while Greece, Ireland, and Portuguese together accounted for 6 percent. Lithuania joined the group at the beginning of 2015, contributing to 0.4 percent of the 19-country currency union.

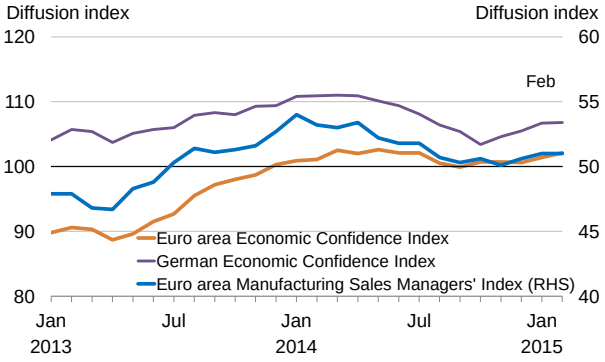
recorded in the previous quarter. Improvements in Germany and Spain from gains in private consumption and exports, which resulted from low oil prices and weaker euro. The Italian economy stalled from the previous quarter, while France's growth moderated due to weakening domestic consumption.

The euro area economy will likely pick up in 2015 Q1 in line with various confidence indices, such as Consumer Confidence Index, European Commission Sentiment Index, and Purchasing Managers' index, which edged up during the first two months of 2015 **(Chart 2.2).** Nonetheless, further risks stem from a probable Greek default, should Greece and its creditors fail to strike a deal for the ongoing debt wrangle, as well as potential escalation of the Russia-Ukraine conflict. Despite the uncertainties, bank credit began to show signs of improvement, as reflected in better credit growth in 2015 Q1 according to the Bank Lending Survey. Such condition should help enhance the ECB's policy transmission to the real economy[2/].

Driven by private consumption and exports, the Japanese economy recovered from a technical recession in 2014 Q4.

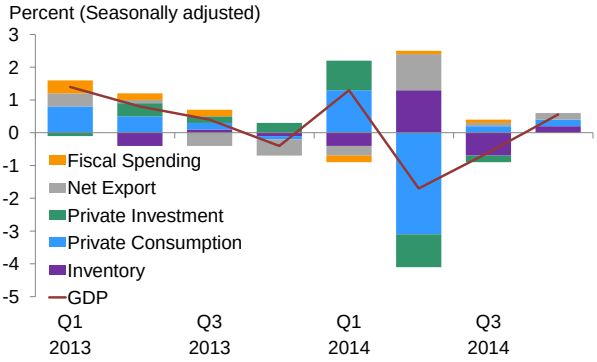
The Japanese Economy in 2014 Q4 expanded by 0.6 percent (qoq sa) **(Chart 2.3)**, after the previous two quarters of contraction. Private consumption improved gradually, while

Chart 2.2 Confidence in the euro area and German economies



Source: Bloomberg

Chart 2.3 Contributions to Japanese Growth (Change from the previous quarter)



Source: Cabinet Office of Japan

[2/] On January 22, 2015, the Governing Council of the European Central Bank agreed to (1) keep all types of interest rates unchanged, (2) expand its asset purchase program to 60 billion euro per month from March 2015 to September 2016, or until the medium-term inflation target of 2 percent is achieved, and (3) cut the interest rate for the TLTROs by 0.1 percent for the remaining six operations.

exports edged up following revived demand from its trading partners. Private investment and fiscal spending, on the other hand, stalled in the last three months of 2014.

Further recovery will likely continue in 2015 Q1. Private consumption is expected to pick up after a strong possibility of wage rise following wage negotiation round, scheduled around mid-year. Also, manufacturing sector will likely improve in line with the rise in domestic demand and exports, with the latter boosted by stronger demands from major trading partners, particularly the U.S. and the UK

At the MPC Meeting on February 18, 2015, the Bank of Japan pledged to maintain the Quantitative and Qualitative Monetary Easing (QQE) measure in order to achieve the 2 percent core inflation target. The Monetary Policy Committee also judged that the economy is on a gradual recovery path, and inflation will likely moderate following the decline in energy prices, although longer-term inflation expectations continued to trend up.

The Chinese and Asian economies (excluding Japan) expanded at a slower pace than expected, with the Chinese growth slowdown adversely affecting exports of Asian trading partners. Nevertheless, the fall in global oil prices helped support consumption in most economies.

The Chinese economy in 2014 Q4 grew at the same rate with the previous quarter, leading to 7.4 percent annual growth, down from 7.7 percent last year. The slowdown was due partly to moderating investment spending. Investment was held down by various fiscal and financial reform measures, including stricter controls over new investment projects in industries with high excess

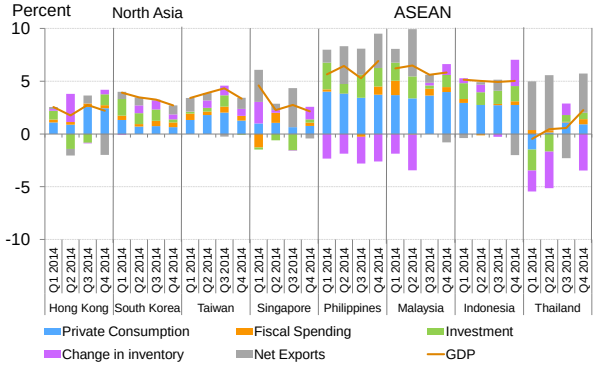
capacity and the tighter supervision of new loan issuance to risky borrowers, particularly in the shadow banking sector. As part of the current reform, the Chinese government has implemented various measures to mitigate the slowdown in growth, including lowering the policy interest rates and bringing forward investment plans in infrastructure.

In 2015 Q1, the Chinese growth will likely remain subdued as investment slows in real estate due to excess supply, and in manufacturing due to falling profit margins. Moreover, local government investment faces limitations from declining land-related revenue following lower land prices. However, private consumption is projected to expand at a rate close to the previous quarter, as labor market conditions remain solid. Exports should also pick up in line with stronger signs of the U.S. recovery.

Given various indicators pointing to weak economic activities and tight liquidity conditions, the People's Bank of China (PBOC) on February 4, 2015, announced a reduction in the Reserve Requirement Ratio (RRR) for all types of financial institutions from 20 to 19.5 percent. Then, on February 28, 2015, the PBOC eased its monetary policy further by cutting both the lending rates and the deposit rates by 0.25 percent, after a similar decision last quarter.

In 2014 Q4, the Asian economies (excluding China and Japan) expanded at a slower pace than in the previous quarter **(Chart 2.4)**, after exports softened due to a fall in Chinese and regional demands. However, consumption growth was robust, thanks to the decline in oil prices that boosted households' disposable incomes. Overall inflation moderated, largely from

Chart 2.4 Contributions to growth of Chinese and Asian economies (Change compared to the previous quarter)



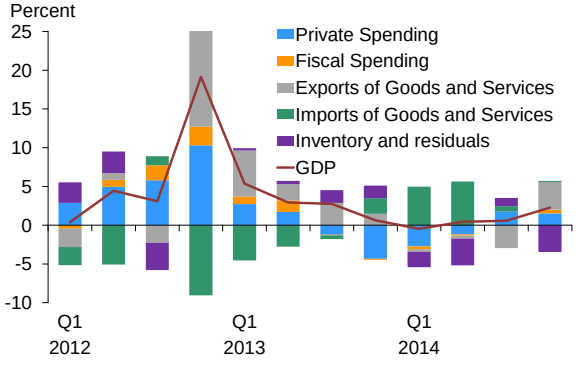
Source: CEIC and Bank of Thailand

the sharp fall in energy and food prices. With inflation in the target range, central banks in the region kept their policy interest rates unchanged, deeming them sufficiently supportive to continued economic growth.

Recent economic indicators in 2015 Q1 suggested moderating growth in the Asian economies, compared to the previous quarter. A number of economies began to witness slowing exports from falling Chinese demand. Weak exports, in turn, have led to a stall in investment. Meanwhile, domestic demands remained subdued, despite lower energy prices.

The sharp fall in global oil prices should be a driver behind global economic recovery by boosting purchasing powers, and consumption. However, such effects may take time to materialize or become diluted by various factors: (1) investments by net exporters of petroleum products, such as Malaysia and Indonesia, are declining; and (2) domestic oil prices in some countries, such as Indonesia, do not fall to the same extent with the drop in global oil prices, because the governments have taken this opportunity to reduce energy subsidies.

Chart 2.5 Contributions to Economic Growth
(Compared to the same period of the previous year)



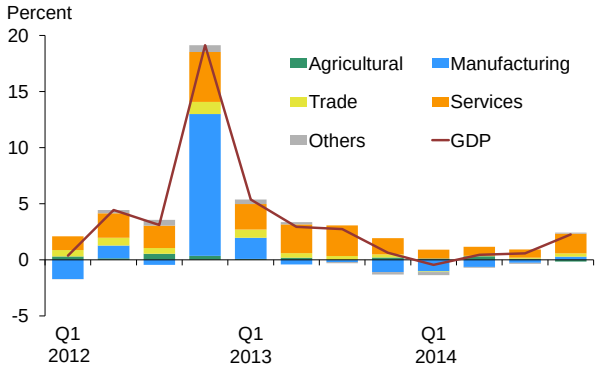
Source: National Economic and Social Development Board

2.2 The domestic economy

The Thai economy continued to recover, albeit at a slower pace than previously assessed. Major drivers of growth in 2014 Q4 were the rise in the number of Asian tourists, who became less concerned about the political situation in Thailand, and larger-than-expected export growth from stronger demand from the U.S. and the CLMV region **(Chart 2.5).**

Meanwhile, domestic production edged up from a mild recovery in manufacturing industries. Agricultural production also improved, but insufficient to compensate for the fall in prices, especially of rubber, leading to a fall in agricultural income **(Chart 2.6).**

Chart 2.6 Contributions to GDP growth
(Change from the same period last year)

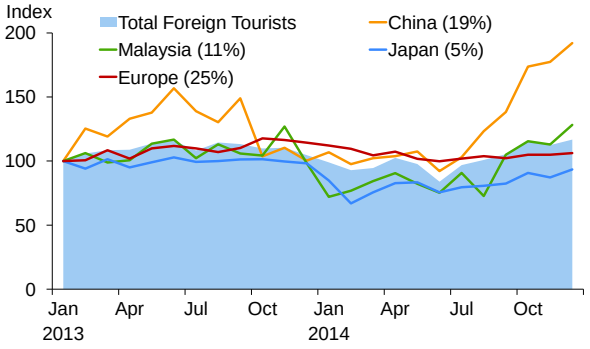


Source: National Economic and Social Development Board

Demand conditions

In 2014 Q4, exports of services, most importantly tourism, resumed their role as a key driver of growth, with a momentum carried over into 2015 Q1. Asian tourists, particularly Chinese and Malaysian, contributed to the growth due to regained confidence over political situation and various tourism stimulus measures at the beginning of 2014 Q4[3/] **(Chart 2.7)**. The increase in number of Asian tourists offset the fall from other major markets, including Europe (including Russia) and Japan, whose economies experienced sluggish growth, currency depreciations, as well as official warnings against travel to Thailand.

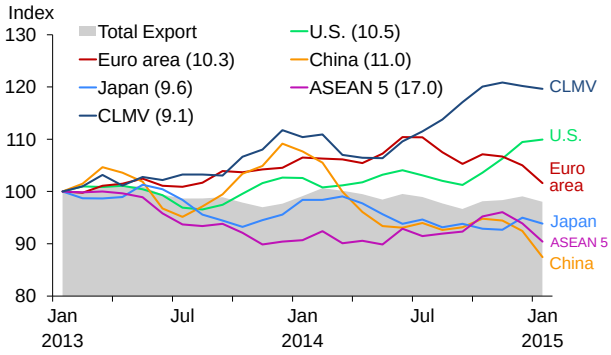
Chart 2.7 Number of tourists by origins
(Indices, January 2013 = 100)



Note: Parentheses ( ) indicate shares of total foreign tourists in 2014
Source: Department of Tourism

In 2014 Q4, export quantity expanded due to recovery in the U.S. orders and robust CLMV demand **(Chart 2.8)**, leading to gains in exports of many industries, including machinery and equipment, petrochemical, electronics, automotive, and agriculture, particularly rice and sugar. Also, European importers expanded their orders before the end of GSP in 2015. Imports, on the other hand, contracted substantially on the sharp decline in commodity prices, particularly oil. Looking ahead, in 2015 Q1 exports are likely to recover at a gradual pace, given the slowdown in the Chinese and ASEAN economies. Moreover, Thai

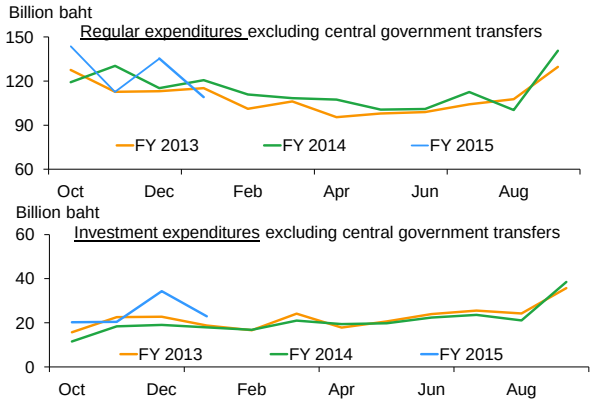
Chart 2.8 Export value indices by trading partners
(Seasonally adjusted, 3-month moving average) (Jan 2013 = 100)



Source: Customs Department (Calculations by Bank of Thailand)

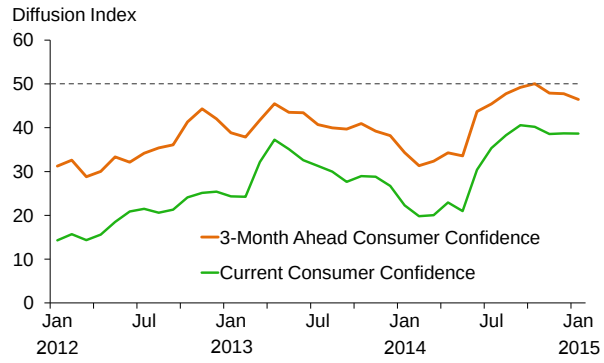
[3/] Visa exemption for Chinese tourists and permission for private vehicles from Malaysia to travel beyond the Songkhla province.

Chart 2.9 Public Expenditures



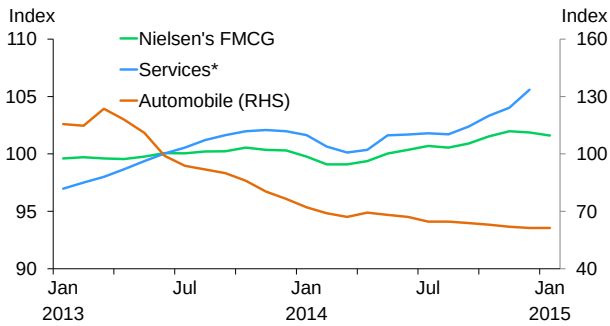
Source: Comptroller General's Department, Fiscal Policy Office

Chart 2.10 Consumer Confidence Index



Source: Ministry of Commerce, calculations by Bank of Thailand

Chart 2.11 Private Consumption Indicators
(seasonally adjusted, 3-month moving average) (2013 = 100)



Note: *Consists of Hotels and Restaurants VATs and Sales of transportation services
Source: Nielsen, The Federation of Thai Industries, The Revenue Department, Calculations by Bank of Thailand

manufacturers still suffered from a lack of technological capabilities required to adapt to the changing global demands. Such limitation was evident in the exports of Hard Disk Drive, which is growing at a slower pace compared with those of regional competitors. Further headwinds include falling commodity prices, most importantly oil prices, which will hurt some export values going forward. (See Box "Impacts of China's economic slowdown on the Thai economy.")

In 2014 Q4, fiscal spending increased from the previous quarter **(Chart 2.9)**, although the efficacy of government investment implementation remained limited. Nonetheless, fiscal consumption expenditures saw a slight increase thanks to various measures to speed up budget disbursements. Due to earlier fiscal spending, public expenditures in 2015 Q1 are projected to moderate. Another drag on fiscal spending will stem from the delay in the government procurement process, as the auction floor prices were revised down following the drop in oil prices. (See Box "Challenges in public investment budget disbursement.")

In 2014 Q4, private investment expanded at a slower rate, and is projected to continue its pace in 2015 Q1. The projected subdued growth is a result of (1) weak farm incomes following declining agricultural prices, (2) elevated level of household debt, (3) cautious lending behavior of financial institutions, and (4) concerns over future income, reflected by weakening consumer confidence **(Chart 2.10)**. The four factors together rendered consumers more careful about spending **(Chart 2.11)**.

Private investment grew at a slow pace in 2014 Q4, a trend which is likely continue in 2015 Q1. The business sector delayed new investment

projects, awaiting greater clarity on economic recovery and government infrastructure investment projects **(Chart 2.12)**.

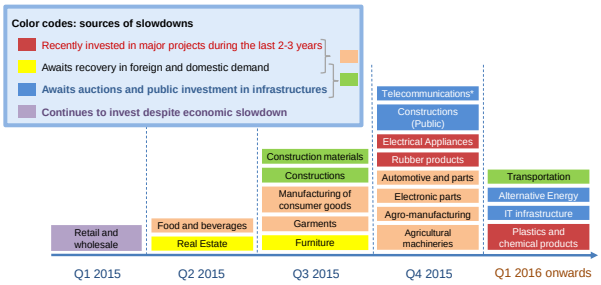
Imports in 2014 Q4 remained at a low level, and are unlikely to pick up in 2015 Q1, in line with sluggish private consumption and investment, as well as slow recovery in exports. Import contraction was a result of the sharp decline in commodity prices, especially oil.

Supply conditions

Industrial production posted a mild growth in 2014 Q4, after six consecutive quarters of contraction **(Chart 2.13)**. Greater demands from the U.S. and CLMV, as well as acceleration of exports to the euro area before the expiration of GSP, were largely responsible for the expansion. This was also reflected by the increase in electricity utilization, the larger number of workers in industrial sector, and a rise in imports of raw materials (excluding fuels and chemical products). In 2015 Q1, industrial production is projected to recover only gradually, as a result of slowing demand for exports from China, ASEAN, and the euro area. In addition, domestic demand growth is likely to remain weak, given lower-than-average domestic spending.

The service sector expanded from a rise in the number of tourists, which grew by 12.8 percent in 2014 Q4. The improvement is likely to continue in 2015 Q1, as tourists, especially Chinese and Malaysian, become less concerned about the political situation. Tourism industry, especially hospitality, will continue to benefit from the development **(Chart 2.14)**.

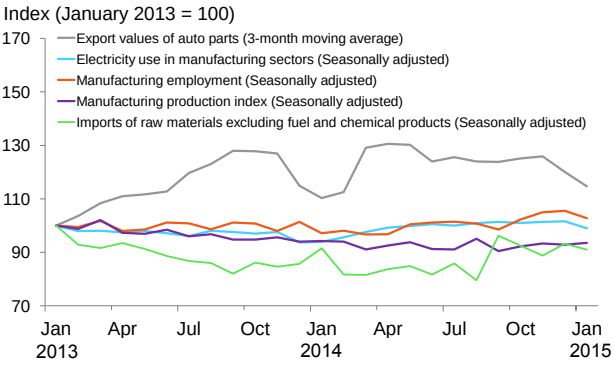
Chart 2.12 Factors that influence private investment decisions



Note: Investment delayed from Q3 to Q4 due to the delay in 4G auction from early Q3 to end of Q3 (September).

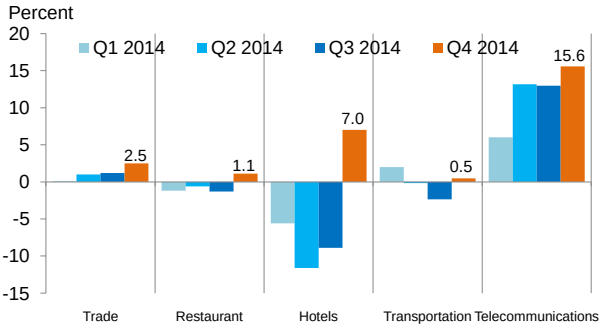
Source: Interviews of business owners from December 2014 to February 2015, Analysis by Economic Intelligence Team, Bank of Thailand

Chart 2.13 Manufacturing production



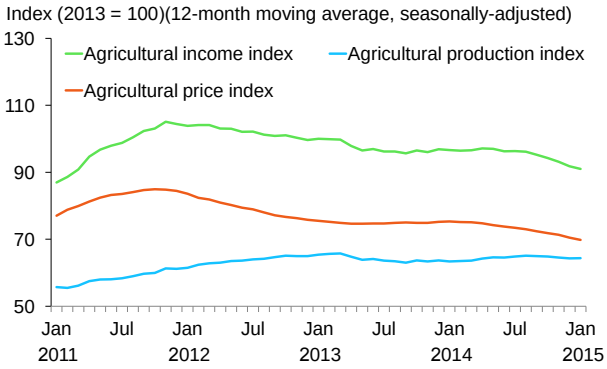
Source: Office of Industrial Economics, Ministry of Industry, National Statistical Office, The Customs Department, The Thai Automotive Industry Association, and the Bank of Thailand

Chart 2.14 Growth of the service sectors



Source: National Economic and Social Development Board

Chart 2.15 Agricultural income and production



Source: Office of Agricultural Economics and Bank of Thailand

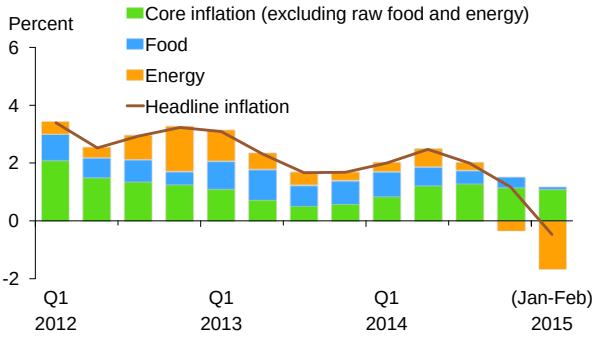
Construction and real estate activities picked up in 2014 Q4, as the number of new home loans rose. Nevertheless, in 2015 Q1 the industry's growth are projected to decelerate slightly, due to cautious spending behavior among consumers. Agricultural production contracted in 2014 Q4, especially after the decline in rubber price had disincentivized rubber tapping. In 2015 Q1, however, agricultural production is expected to rebound, due to the government's buffer stock scheme for rubber, which will encourage more rubber tapping, as well as favorable weather conditions. In addition, production of shrimps, which has suffered from an epidemic over the past two years, is likely to recover in 2015. However, the improvement in rubber and shrimp production will be insufficient to compensate for the subdued agricultural prices, especially for rubber, leading to a fall in overall farm incomes **(Chart 2.15)**.

Employment in 2014 Q4 were largely stable. Non-agricultural employment rose as a result of exports recovery, as well as labor migration from agricultural to non-agricultural sectors, following the weak agricultural prices. Moreover, the growth of non-agricultural wages indicated expanding demand for labor in these sectors. However, in 2015 Q1, unemployment rate is expected to rise slightly due to the fall in farm employment, although some of these workers will enter into industrial and construction employment.

2.3 Production costs and prices conditions

Headline inflation is expected to decline, consistent with the previous assessment and global inflation trend. The plunge in oil prices, driven by excess supply of global crude oil, contributed to a fall in production costs and hence lower inflation readings. However, current conditions do not indicate deflation risks, given the continued rise in the prices of most goods and services and inflation expectations remaining close to target **(Chart 2.16).**

Chart 2.16 Contributions to headline inflation



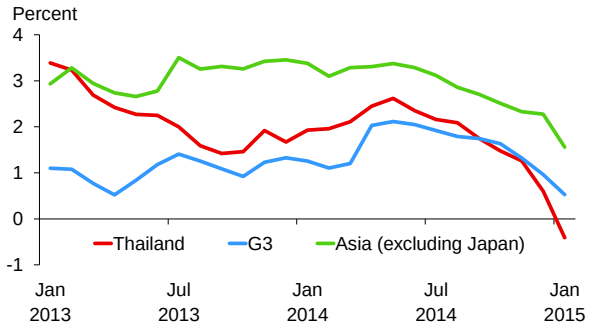
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Quarterly inflation

Unit: Percent	2013	2014				2015
		Q1	Q2	Q3	Q4	Jan-Feb
Percentage change from previous year (%Δyoy)						
- Headline Consumer Price Index	2.18	2.00	2.47	2.00	1.11	-0.47
• Core Consumer Price Index	1.00	1.19	1.70	1.79	1.65	1.54
• Raw food	5.54	5.34	3.91	2.78	1.88	0.51
• Energy	4.79	2.55	5.13	2.20	-3.11	-13.39
Percentage change from previous quarter (%Δqoq)						
- Headline Consumer Price Index	-	0.8	0.9	-0.1	-0.5	-
• Core Consumer Price Index	-	0.5	0.7	0.2	0.2	-
• Raw food	-	0.7	2.2	-1.0	-0.1	-
• Energy	-	2.3	1.0	-1.1	-5.2	-
Percentage change from previous quarter (%Δqoq_sa)						
- Headline Consumer Price Index	-	0.8	0.5	0.0	-0.2	-
• Core Consumer Price Index	-	0.5	0.6	0.3	0.2	-
• Raw food	-	1.9	0.1	-0.6	0.5	-
• Energy	-	0.5	0.6	-0.7	-3.5	-

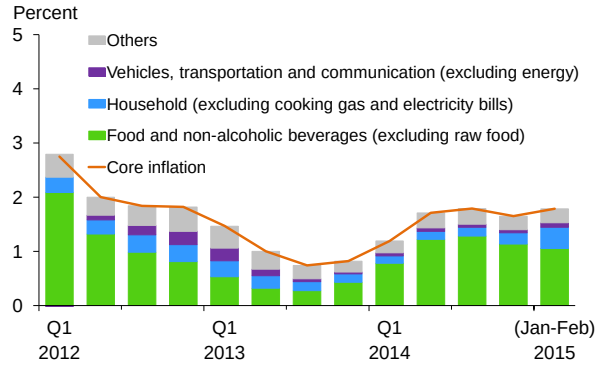
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and seasonally adjusted quarter-on-quarter percentage change calculations by Bank of Thailand.

Chart 2.17 Domestic and foreign headline inflation



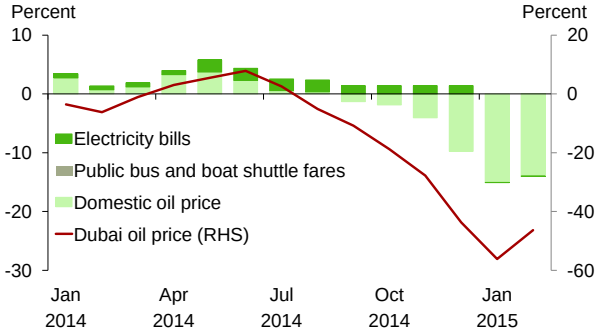
Notes: Foreign headline inflation calculated from unweighted averages of individual countries inflation
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, CEIC, and calculation by Bank of Thailand

Chart 2.18 Contributions to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.19 Changes in prices of goods and services following the fall in oil price



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Since 2014 Q4, global commodity prices, particularly crude oil prices, have declined massively, leading to falling domestic energy prices. As a result, headline inflation fell to a negative 0.47 percent in the first two months this year, in line with global inflation trend **(Chart 2.17)**. Core inflation, on the other hand, remained stable, averaging at 1.54 percent for the same two-month period **(Table 2.1 and Chart 2.18)**.

The fall in oil prices affected domestic prices of goods and services via two main channels. (1) Direct impact on domestic energy prices, which fell by 21.75 percent on average over the first two months of 2015.[4/] (2) Indirect impact on production and transportation costs, including lower electricity bills, reduced fares for first and second-class air-conditioned public buses, as well as boats **(Chart 2.19)**. Despite the fall in production costs, prices of most goods and services have not been marked down noticeably, since firms normally cut prices when the reduction in costs is deemed sustained and substantial. Also, since transportation costs only account for about 2.29 percent of GDP, the impact of lower production costs has been marginal[5/].

The decline in headline inflation can be largely attributed to the substantial fall in oil prices, rather than to sluggish demand. In addition, negative inflation reading does not give rise to deflation concerns, because (1) prices of most goods and services, except energy-related, remain close to long-term historical averages, and (2) one-year

[4/] Over the first two months of 2015 Q1, the prices of diesel and gasohol 91 fell by 13 and 29.31 percent, respectively.

[5/] Information from the presentation titled “Assessing the effectiveness of logistics in the industrial sector” by the Ministry of Industry.

ahead inflation expectations, according to Consensus Economics and Bank of Thailand's Business Sentiment Survey, are 2.10 and 2.58 percent respectively **(Chart 2.20).** This points to the public's assessment that negative headline inflation was transitory in nature.

Given subdued global commodity prices, especially crude oil prices, inflationary pressures are likely to remain low for the next two quarters. Although global oil prices rebounded moderately in February 2015, excess oil supplies, coupled with weak global demands, should keep the prices at low levels for an extended period. Meanwhile, slower-than-forecast recovery of the Thai economy will exert additional downward pressure on short-term inflation, consistent with Underlying Inflation,[6/] which indicated low inflationary pressures in the first half of 2015 **(Chart 2.21)**.

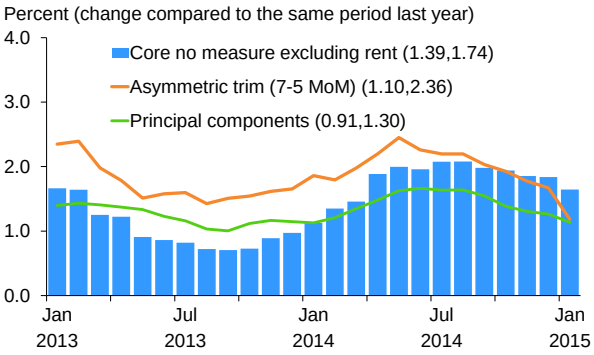
Looking ahead, headline inflation is expected to remain negative for some time, given low energy prices. Our assumptions of a recovery in oil prices, from lower supplies, and a gradual global economic recovery led to our projections of positive headline inflation in the second half of this year.

Chart 2.20 Short-run inflation expectation (one-year ahead)



Source: Business Sentiment Index Survey by Bank of Thailand in January 2015, and inflation expectation of economists surveyed by Consensus Economics Inc. as of March 2015.

Chart 2.21 Underlying Inflation



Notes: Inside parentheses, figures on the left indicate February 2015 %YoY, while those on the right show 2010-2014 average.

[6/] Please refer to FAQ issue 93 "Indicators for inflation trend" in www.bot.or.th

Impacts of China's economic slowdown on the Thai economy

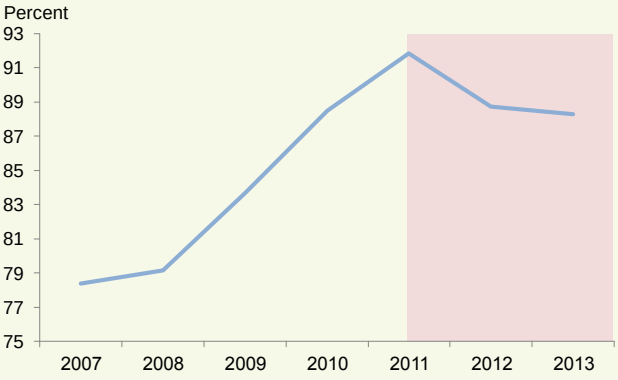
Since 2010, China's economic growth has been decelerating following the period of high economic growth. While it is natural for an economy to experience slower growth as it grows larger, the current slowdown of the Chinese economy could also be attributed to the Chinese government's economic reform. The reform puts a stronger emphasis on stability than growth, and aims at relying on domestic consumption, rather than exports and investment, as the main propeller for growth. Such reform policies have led to a slowdown in domestic demand in the short run and a slowdown in import demand, in particular for commodities **(Chart 1)**.

Chart 1 China's GDP and import growth continue to decline



Source: CEIC and the World Bank

Chart 2 China's imports to exports ratio of electronics products



Source: Trademap calculated by the BOT

Despite moderating domestic demand, China's exports are projected to continue expanding, as the recovery in advanced economies led to growing demand for intermediate goods, which in turn constitutes as much as 27 percent of total imports value. However, **the growth rate of China's imports of such goods may not be as high as it was in the past due to certain structural limitations**. China now places a stronger emphasis on horizontal integration, i.e. increasing domestic production of goods within its production chain as substitutes for imports of specialized goods from various other countries, particularly from ASEAN **(Chart 2)**. There is also increased vertical integration, i.e. trading of intermediate goods. This is a result of China's economic reform policy which aims to upgrade the country's production chain to a more technologically advanced and sophisticated level, and the relocation of production base to other countries in response to the sharp increases in production costs in the Chinese market.

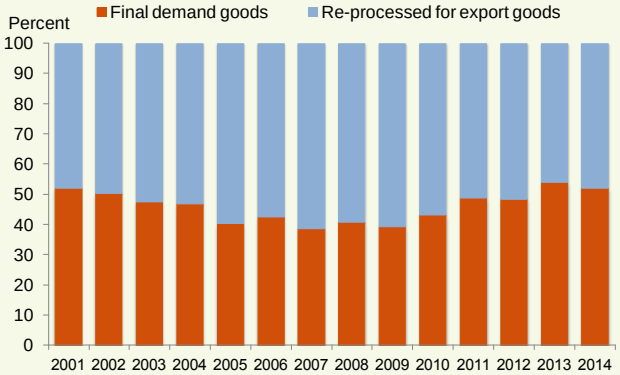
China is the most important market for the Thai economy, both in terms of exports and tourism. The Chinese market constitutes 11 percent of Thailand's total exports, while Chinese tourists account for 19 percent of total visitors to Thailand in 2014. The effects of China's

economic slowdown are expected to be felt largely by the export sector, while the effect on tourism should be marginal.

Impacts on the merchandise exports

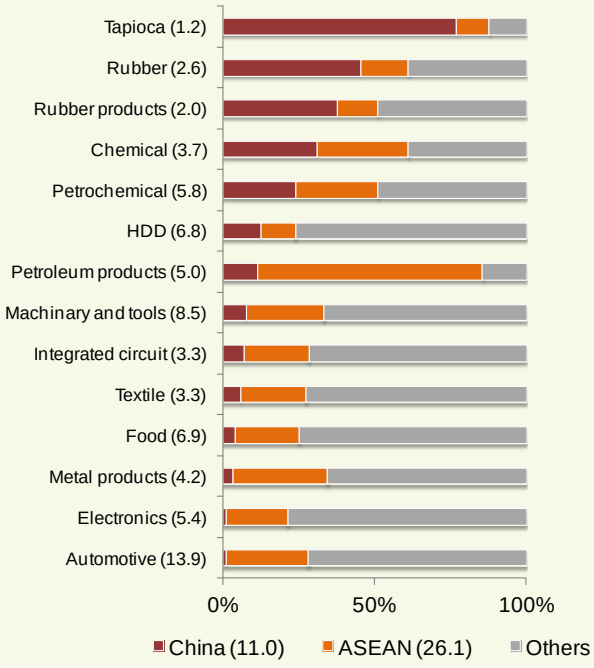
The slowdown of the Chinese economy is likely to affect the Thai export sector more significantly than it did in the past. This is because it is an outcome of a structural change in the Chinese economy, which in turn caused a moderation of domestic demand during the reform period, as well as a decrease in the demand for goods used for domestic consumption (final demand)[1/]. The effect of the slowdown on the Thai economy will be more pronounced than in the past because Thai exports of final demand goods constitutes more than 50 percent of total Thai exports to China in 2014. This contrasts with the situation in the past where a major share of Thai exports to China constituted goods which were re-processed for exports[2/] **(Chart 3)**. The shift in the export composition is due to the following factors: (1) The Chinese government's policy to stimulate the economy after the global financial crisis, making China the world's largest consumer of several commodities, (2) China's economic reform which places a strong emphasis on developing the manufacturing sector, making China less reliant on imported intermediate goods, and (3) The global shift in consumer preference away from desktop computers and notebooks towards tablets and smartphones, resulting in a significant

Chart 3 Ratio of Thailand's exports to China, by objectives



Source: Customs Department and calculations by the BOT

Chart 4 Ratio of Thai exports to China and to ASEAN (ratio of total Thai exports)



Source: Customs Department and calculations by the BOT

[1/] Goods that are used for domestic consumption in China, rather than for re-processing for export. These goods include tapioca, rubber and rubber products, processed food, petroleum products, etc.

[2/] Goods in the re-processed for export category include HDD, IC, chemicals, petrochemicals, machineries and tools, etc.

decline in Thai exports of HDD—a major component of Thailand's re-processing for export category.

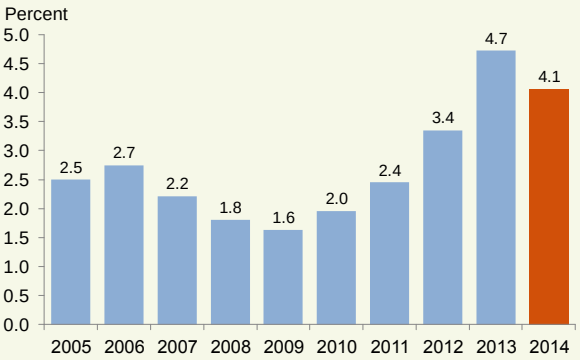
Therefore, the economic slowdown in China is likely to weigh on Thailand's exports volume of final demand goods, especially commodities such as rubber and rubber products, tapioca and petroleum **(Chart 4)**. Moreover, as China is one of the world's major consumer market, its subdued domestic demand would also exert pressure on the global commodity prices, which in turn depress Thailand's farm incomes even in the absence of direct export linkage. However, the exports of intermediate goods should continue to grow in line with China's projected export growth.

On top of the expected slowdown of Thai exports to China, the export sector may also experience an indirect effect through the slowdown of Thai exports to ASEAN. Thai exports to ASEAN makes up as much as 26.1 percent of total exports in 2014. Many ASEAN countries export a considerable amount of consumer goods to China. Therefore, the demand for Thai imports in ASEAN markets may also subside following ASEAN countries' reduced exports revenue. Among Thailand's exports to ASEAN, the items which are likely to feel the impact include automobiles, electronics, petroleum and petrochemicals.

Impacts on the tourism sector

The impact of China's economic slowdown on the tourism industry is likely to be limited, due to the following factors: **1) Demand for foreign travels among Chinese tourists should continue to rise** in tandem with increasing per capita income. Moreover, the share of outbound Chinese tourists to its total population is also likely to increase, although it is still relatively low compared to the developed countries, reflecting ample untapped potential in the tourism sector; **2) Thailand is among the top destinations for Chinese tourists** as it offers value for money. Also, Thailand can be easily reached, now that several airlines have started operating new routes, facilitating the transportation of tourists from various Chinese provinces to Thailand. Thailand's share as a destination for outbound Chinese tourists is therefore constantly on the rise. However, the political unrest in Thailand in 2014 caused a drop in Thailand's share from the previous year **(Chart 5)**. Thus, given that the political situation in Thailand remains stable in 2015, the share should gradually pick up; **3) The protest against the Chinese government and Chinese tourists in Hong Kong should benefit Thai tourism, as some Chinese tourists are likely to shift their destination from Hong Kong to other countries** (Hong Kong has the highest share of Chinese tourists at 40 percent). This can be reflected from the

Chart 5 Thailand's market share as a destination for China's outbound tourists



Source: Tourism Authority of Thailand and China National Tourism Administration

February 2015 statistics during which there was a long public holiday in China. At the time, Thailand received an increasing number of Chinese tourists, while the number of Chinese tourists to Hong Kong contracted for the first time in 20 years.

From the above factors, **the number of Chinese tourists coming to Thailand is projected to grow solidly despite the expected economic slowdown. The amount of spending per day by Chinese tourists is also unlikely to be affected,** as the slowdown in the recent period had in fact corresponded with higher daily spending of Chinese tourists, while their stay continued to lengthen **(Chart 6)**. In the long run, however, the policy should focus on the revenues received from tourism rather than the number of tourists. Historically, price has been a key selling point of Thai tourism. In the longer run, however, this could become an obstacle for the industry to upgrade itself to a higher quality and value-added level, and therefore hold back its revenue potential in the periods ahead.

Chart 6 Amount of daily spending and length of stay for Chinese tourists



Source: Survey by Department of Tourism

In summary, China's economic slowdown that resulted from its reforms is likely to weigh on Thai exports, especially merchandise exports for domestic consumption in China. It may also indirectly affect Thai exports through moderating exports to ASEAN. Exports of intermediate goods used for re-processing for exports, while the tourism sector, should be relatively unaffected. The slowdown of the Chinese economy and ASEAN is one of the reasons for the BOT's downward revision of export value growth in 2015 to 0.8 percent, compared to 1.0 percent in the December 2014 report.

Challenges in public investment budget disbursement

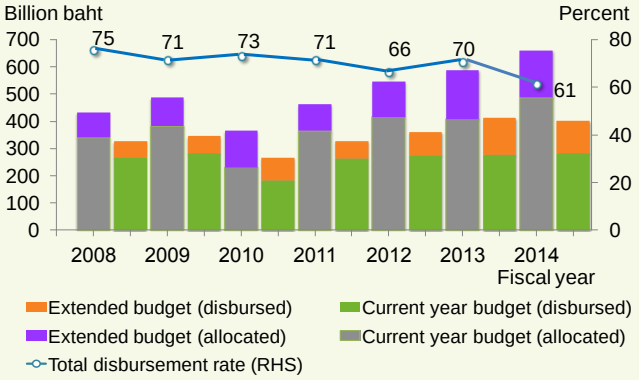
For several years, the government has recognized the importance of public investment in lifting the country's economic potential. Larger funds have therefore been allocated for public investment, both in the annual budget and fiscal stimulus packages in some years. Despite the government's attempts in expediting budget disbursements, aggregate fiscal injection into the economy has not increased as much as hoped. This raises an important question as to what exactly impedes budget disbursement, and will they be resolved in the future?

1. **Public investment** is affected by two primary factors.

(1) Structural problems stemming from limitations of the efficiency of public agencies

In the past five years, the government allocated increasingly larger funds for public investment. Public agencies therefore had to execute greater number of investment projects simultaneously. However, inefficiencies in the operations of government agencies resulted in a gradual downward trend in public investment disbursement, while the increase in public spending on investment has turned out to undershoot expectations **(Chart 1)**.

Chart 1 Budget disbursements by public agencies



Source: GFMIS and calculations by the Bank of Thailand

Analysis of the trend in public investment budget disbursement reveals important facts. The delayed passage of the budget bill for fiscal year 2012 caused most public agencies to postpone their investment plans, with the allocated funds accumulated for subsequent years. Public agencies also have had to implement projects that drew on extra-budgetary funds. For example, the remaining funds from "Thai Khem Khaeng" fiscal stimulus were used to finance the housing repair projects in fiscal year 2015. With constraints on their operations, public agencies struggled to cope with the sheer increase in the number of government investment projects. The restrictions stemmed from the low capacity caused by inadequate number of staff and a shortage of capable contractors to undertake projects in provincial areas. Furthermore, the government's auction floor prices for construction projects were sometimes lower than the current market rate, discouraging firms from participating in project auctions. This caused delays in the procurement for some public agencies. Nevertheless, since early 2015 the government has attempted to expedite budget disbursement by urging public agencies to enter into contracts of investment projects. At the end of February 2015, government agencies signed contracts for investment projects accounting for a sizable 30 percent of the total public investment expenditure

– a fairly high figure compared to those of the previous years. Nevertheless, due to public agencies' efficiency constraints, the rate of disbursements for public investment over the first five months of fiscal year 2015 remained modest.

(2) Temporary factors: political situation and lower auction floor prices of government construction projects

(2.1) Political situation. Political uncertainty contributed to much of the delay in budget disbursements for public investment. Since the easing of the political situation in 2014 Q3, the National Council for Peace and Order (NCPO) has urged government agencies to expedite budget disbursement for public investment and enter into contracts on investment projects. However, these measures have not yielded satisfactory outcome because of the failure to resolve the structural constraints discussed above. In addition, if government agencies fail to enter into contracts before the deadline, they can request their investment projects to be undertaken at a later time, further causing the delays in budget disbursement.

(2.2) Measure to reduce the auction floor prices of government construction projects was introduced by the Cabinet on December 30, 2014, on the back of sharply lower diesel prices. The measure affected government agencies that had been in the process of procurement or implementation of investment projects (before December 16, 2014) but had not signed the contracts. These agencies would have to lower the construction costs according to the new auction floor rates. This measure is expected to cause some investment projects to be delayed by two months, but will not have a substantial impact on budget disbursements.

2. **State enterprise investment,** which accounts for almost half of public investment, has also faced delays in disbursement in recent periods due to two main factors.

(1) Uncertainty in government policy and the country's long-term investment plan

Several state enterprise investment projects experienced delays as a result of uncertainty in government policy and long-term investment plan. Investment projects and long-term financing plans were altered by each different government. For example, the high-speed train project or the 2-trillion-baht infrastructure development bill was ruled unconstitutional. This prompted the government to revert to the normal budget allocation procedure where loans extended to state enterprises are reviewed on a case-by-case basis.

(2) Policy to address structural problems of state enterprises

(2.1) Anti-corruption measures. The NCPO established the Committee on the Inspection of Public Expenditures to ensure transparency in contracts and procurement concerning large-scale public investment projects. As a result, several projects are being reviewed and some were canceled.

(2.2) State enterprise reform. The NCPO set up the State Enterprise Policy and Supervisory Committee tasked with formulating policy and supervisory standards for state enterprises. The goal was to increase efficiency in state enterprise operations and quality of

service. Construction Sector Transparency Initiative and Integrity Pact standards were used to enhance transparency in every step of government procurement. On February 24, 2015, the Committee also devised and approved restructuring plans for state enterprises with heavy losses, prompting these state enterprises to review their previous investment plans.

The anti-corruption and state enterprise reform measures may have delayed the disbursements for public investment in the short term, but over the long term should help enhance the transparency of the procurement process and reduce the accumulated losses of state enterprises. The measures are also expected to lead to greater efficiency in the state agencies and better returns to public investment.

Beside the above two factors, **state enterprises also face varying structural problems.** For example, the PTT group needs to cope with lower global oil prices and delays in the latest round of petroleum concessions, which are awaiting legal amendments. The Electricity Generating Authority of Thailand (EGAT) suffers from a shortage of technicians, and a lengthy process of investment project approvals caused by the need to evaluate the diverse impacts of the projects. It takes considerable time to address these structural problems.

In conclusion, the government has made an attempt to expedite public investment and address short-term limitations to budget disbursements. Such measures include the monthly monitoring of disbursements of individual state agencies, and the Budget Bureau's lifting of auction floor construction prices for some agencies. The government also aims to overcome long-term impediments to budget disbursements. To reduce uncertainty in government policy, clearer and consistent long-term investment planning are devised through the implementation of priority investment projects and state enterprise reform. However, these efforts may take time to materialize, especially with regard to the limited efficiency of the state agencies. If the government is able to prioritize investment projects, allocate sufficient number of staff in line with increasing investment projects, and cut back on new regulations or measures that may hamper budget disbursements for investment projects, then these ongoing efforts will be able to enhance the government's role in driving the economy, as well as boosting private sector confidence and private investment. Going forward, higher overall investment will contribute to raising Thailand's economic potential.

Monetary and Financial Stability

3. Monetary and Financial Stability

Thailand's overall economic and financial stability was well maintained. Monetary conditions remained accommodative and supportive of economic growth. Stability of financial institutions were stable and continued to maintain previous lending standards, while stability of non-financial institutions strenghtened following economic recovery in 2014 Q4.

The Monetary Policy Committee deemed it necessary to monitor the following risk factors going forward. (1) Additional monetary policy easing from major and regional economies, which may exert upward pressures on the Thai baht, (2) Recovery paths of income and loan repayment abilities, especially among high-risk groups such as SMEs, low-income households and agricultural households, whose earnings were depressed due to falling agricultural prices, and (3) Search-for-yield behavior among investors, encouraged by a prolonged period of low interest rates, which could lead to the build-up of financial fragility.

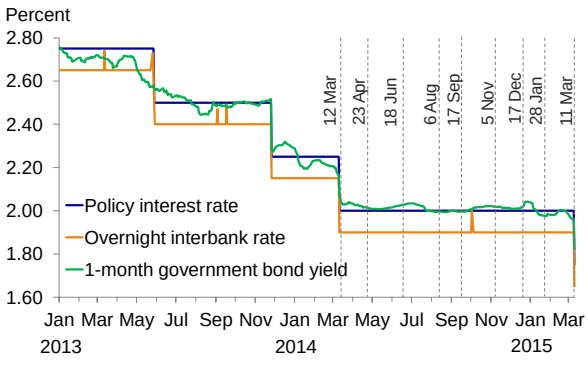
3.1 Financial markets

Overall monetary conditions remained accommodative. Money market interest rates remained low, in line with the policy interest rate, while government bond yields declined over the same period.

Money and bond markets

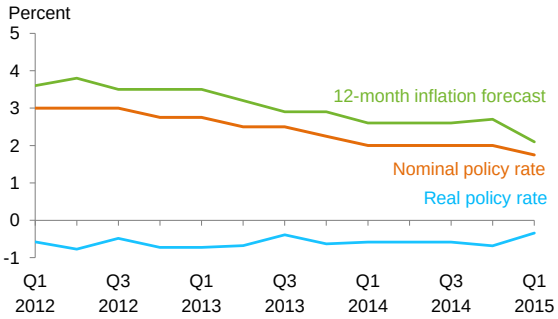
Overall monetary conditions continued to be accommodative and supportive of economic recovery, as reflected by low levels of short-term interest rates, close to the policy interest rate. The MPC maintained the policy rate at 2.00 percent at the January 28, 2015 meeting, before lowering it to 1.75 percent on March 11, 2015 (**Chart 3.1**). Moreover, real policy rate (nominal policy rate subtracted by 12-month-ahead inflation consensus forecast) remained negative, although it edged up

Chart 3.1 Money market interest rates



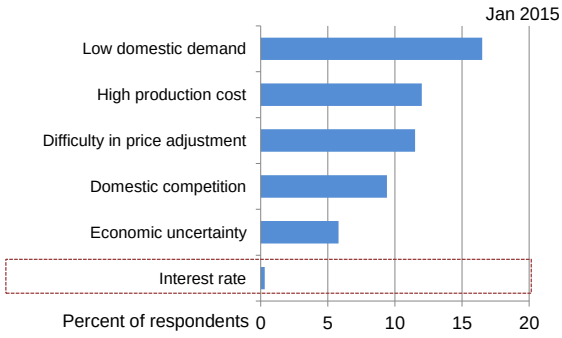
Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.2 Real policy rate*



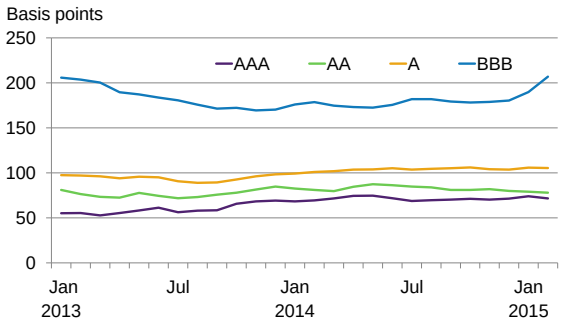
Note: *Real policy rate is calculated by subtracting the 12-month inflation consensus forecast (Consensus Economics Inc.)
Source: Bank of Thailand

Chart 3.3 Impediments to business operations*



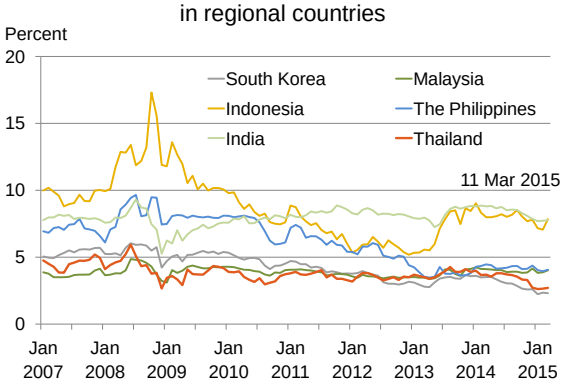
Note: *Rated as number 1 restriction in business operations
Source: Bank of Thailand's survey of business sentiment

Chart 3.4 Credit spread of Thai corporate bonds*



Note: *The difference between corporate and government bond yields (maturity of 3-5 years)
Source: ThaiBMA

Chart 3.5 10-year government bond yields in regional countries



Source: Bloomberg

slightly from the previous period due to the decline in inflation forecast **(Chart 3.2 and Box "Understanding the Real Policy Rate").** Bank of Thailand's survey of business sentiment also indicated that most businesses did not consider the current level of interest rates a major impediment to their business operations **(Chart 3.3).** Likewise, the costs of borrowing in the financial markets remained low **(Chart 3.4).**

Thai government bond yields have declined substantially since December 2014, as a result of mounting concerns over the recovery of the Thai economy as well as the fragility of the global economy. The fall in yields could also be attributed to lower-than-expected inflation as well as the growing demand for Thai bonds by both domestic and foreign investors. As of January 14, 2015, the 10-year government bond yield closed at 2.49 percent, the lowest level in six years.

The government bond yields rose somewhat between the end of January and mid-February 2015. This was in line with the U.S. treasury yields, which climbed as the U.S. economy and labor market showed steady signs of improvement, prompting speculation of an earlier-than-expected hike in the federal funds target rate. The rise in yields also resulted from the offloading of Thai bonds, especially in the 5 to 15-year sector, by domestic and foreign investors due to profit taking and concerns over the Greek debt negotiation. However yields declined slightly following the policy rate reductionon March 11, 2015, and also some rollover demands. On March 11, 2015, the government bond yields declined by approximately 1-15 bps from the previous day's close, with the 10-year yield at 2.74 percent, nearly the lowest level in the region. **(Chart 3.5).**

Equity market

Return-seeking behavior had intensified among equity investors, but risks to the financial system remained contained.

Given accommodative monetary conditions and low level of interest rates, investors sought higher returns both in the Stock Exchange of Thailand (SET) and in the Market for Alternative Investment (MAI). The SET and the MAI indices, therefore, have been on the upward trends since the end of 2014, driven mainly by purchases of domestic institutional investors in the SET **(Chart 3.6)** and of retail investors in the MAI **(Chart 3.7)**. Although the SET and MAI indices, as well as the Forward P/E (Forward Price-Earnings Ratio), declined slightly between March 1-11, 2015, the indices remained elevated and the forward price-earnings ratio higher than 10-year historical average **(Table 3.1)**. These indicators signaled that mispricing of risks among investors may be developing

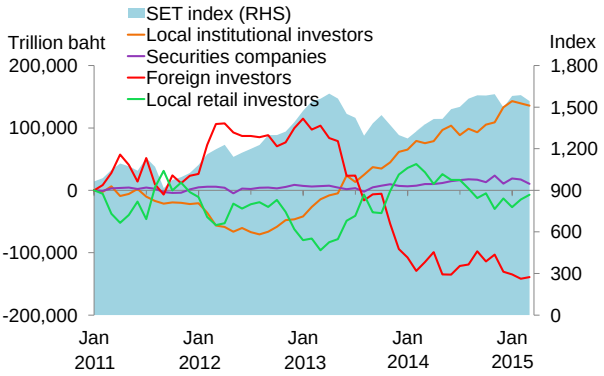
Table 3.1 Forward P/E ratio* in the SET and MAI (times)

Forward P/E Ratio (times)	10-year Average	2014 Q1	2014 Q2	2014 Q3	2014 Q4	2015 Jan	2015 Feb	2015 1-11 Mar
SET	12.0	12.3	13.8	15.5	16.2	14.4	15.4	15.1
MAI	13.8	21.7	26.7	37.2	35.9	20.7	23.0	23.4

Note: Forward P/E Ratio is current price divided by 1-year expected earning per share
Source: Bloomberg and calculations by the Bank of Thailand

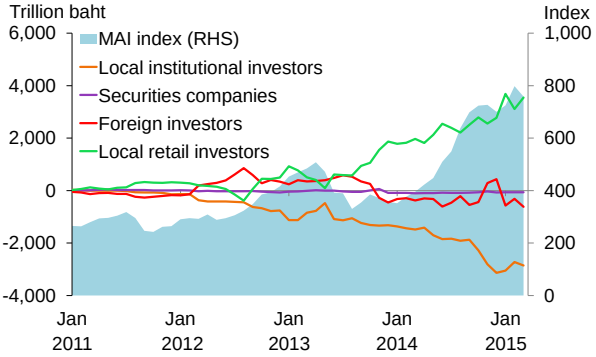
It is worth noting that the amount of credit extended by securities companies for investment in the stock market have been increasing significantly since 2014 Q3, in line with trading volumes in the SET **(Chart 3.8).** Nonetheless, most investors still funded their investment through their own savings, and therefore impacts of a sudden stock market decline on domestic financial markets should be limited.

Chart 3.6 The Stock Exchange of Thailand index and net buy classified by investor types



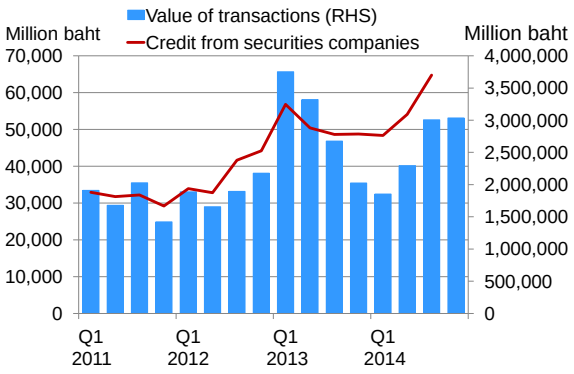
Source: The Stock Exchange of Thailand (latest data on 11 March 2015)

Chart 3.7 The Market for Alternative Investment (MAI) and net buy classified by investor types



Source: The Stock Exchange of Thailand (latest data on 11 March 2015)

Chart 3.8 Borrowing for Stock Market Investment



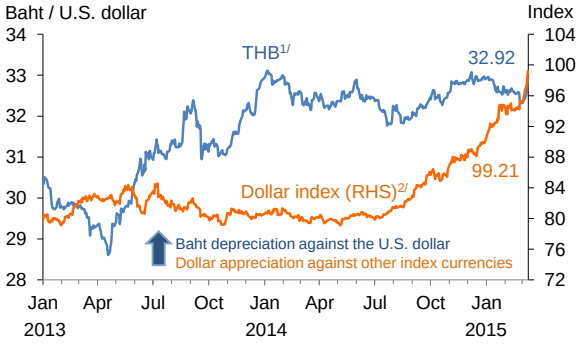
Source: The Bank of Thailand and the Stock Exchange of Thailand

The Securities and Exchange Commission (SEC) and Stock Exchange Thailand (SET) implemented a series of measures, such as Trading Alert List and Turnover List, in order to curb irregular trading activities. These measures led to a decline in net purchases by retail investors in the MAI. Moreover, the SET is currently proposing to the SEC a 1-year silent period measure, on the investors who receive private placement offerings only if the offering prices are below 90 percent of the average price 7-15 days prior to the private placement.

Foreign exchange market

The Thai baht appreciated steadily against the U.S. dollar from January to February 2015, in line with the effective exchange rate indices. The appreciation was largely due to additional monetary policy easings in the euro area and Japan, as well as an improvement in Thailand's current account. However, since March 2015, the baht has depreciated sharply on the back of improving U.S. economic conditions **(Chart 3.9).** With respect to exchange rate volatility, the baht volatility remained relatively low compared to regional peers.

Chart 3.9 Baht-to-U.S. dollar exchange rate and dollar index



Source: 1/Bank of Thailand
2/Bloomberg

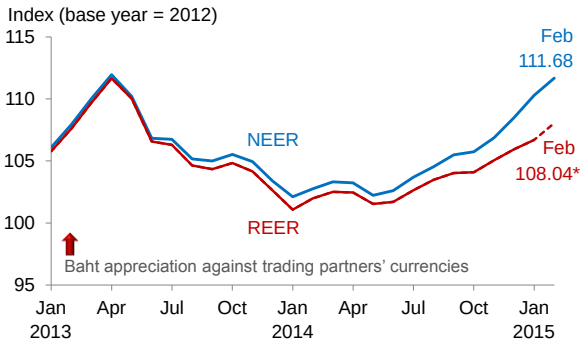
From the beginning of 2015 to February, the Thai baht appreciated against U.S. dollar because of (1) additional monetary policy easings in major economies, particularly the 60-billion euro per month asset purchase plan announced by the European Central Bank on January 22, 2015, and (2) Thailand's larger-than-expected current account surplus from a sharp fall in global oil prices and imports of capital goods **(Box "Thai baht appreciation: its causes, effects, and implications on Thai businesses")**.

In March 2015, however, the baht started to depreciate against the U.S. dollar. The shift was triggered by better-than-expected readings of the U.S. labor market indicators, which further confirmed solid recovery of the U.S economy and increased the expectation that the Fed would raise its policy rate as early as the June meeting 2015. Most recently, after the Monetary Policy Meeting on March 11, 2015, the baht closed at 32.92 baht per U.S. dollar, a 0.1 percent weaker than the end of 2014.

The Nominal Effective Exchange Rate (NEER) soared since the beginning of 2015, reaching 111.68 in February 2015, a 2.9 percent rise since the end of 2014. Consistent with the NEER, preliminary data for the Real Effective Exchange Rate (REER) in February 2015 indicated the index rising to 108.04, suggesting appreciations against most major and regional currencies **(Chart 3.10 and Chart 3.11).**

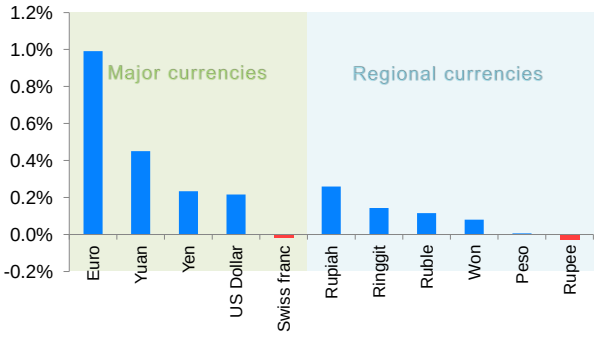
Regarding developments in major currencies, the euro and the yen depreciated as a result of further monetary policy easings, while the Swiss franc appreciated sharply after the lifting of the exchange rate floor for the Swiss franc against the euro. Meanwhile, currencies of oil-exporting countries in the region, such as Indonesia and Malaysia, weakened following a fall in oil exports. The baht, however, exhibited lower volatility compared to regional currencies. **(Chart 3.12)**

Chart 3.10 NEER and REER



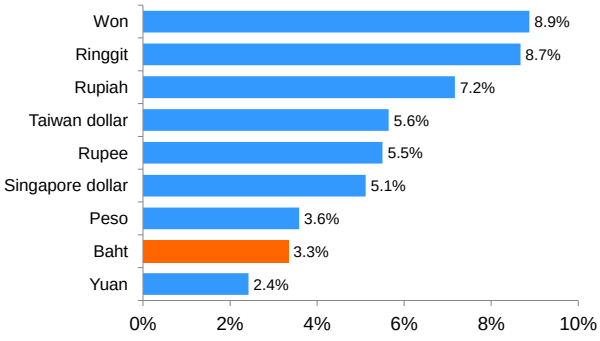
Note: *Preliminary data using NEER in February 2015 and inflation in January 2015
Source: Bank of Thailand

Chart 3.11 Contribution to NEER
(February 2015 compared with December 2014)



Note: + = depreciation against Thai baht
Source: Bank of Thailand and Reuters

Chart 3.12 Exchange rate volatility
(average volalitity: beginning of 2015 – 27 February 2015)



Source: Bank of Thailand and Reuters

3.2 Financial institutions

From December 2014 to February 2015, deposit and loan rates of the four largest commercial banks stabilized from the end of November 2014. On the other hand, credits and deposits improved moderately following gradual recovery of the Thai economy. Banks continued to compete for deposits to maintain their customer base.

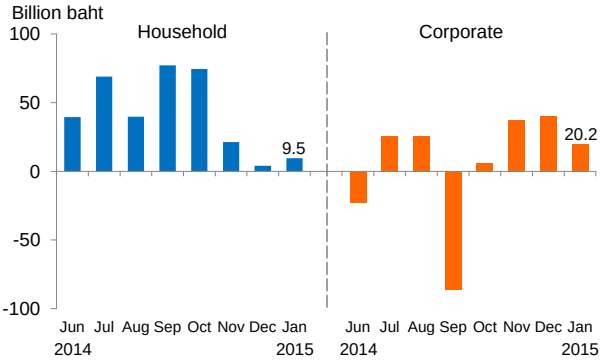
Interest rates, credits and deposits

The average Minimum Lending Rate (MLR) of the four largest commercial banks stood at 6.75 percent at the end of February 2015, unchanged from the end of November 2014.

New issuance of private credits[1/] improved gradually from 2014 Q4 to January 2015 **(Chart 3.13)**, in line with the pace of the economic recovery. Stronger private investment contributed to a moderate rise in business loans. Although small and medium firms faced more stringent lending standards, those for large firms remained unchanged. In addition, some of the larger firms relied on funding from corporate bond and equity issuance with lower funding costs **(Chart 3.14)**.

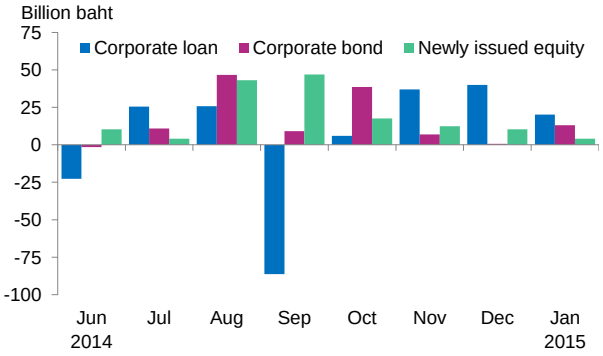
Household credits contracted compared to 2014 Q3, as a result of cautious spending behavior among households, given subdued economic

Chart 3.13 New private credits
(Changes in credit balance, seasonally adjusted)



Source: Bank of Thailand

Chart 3.14 Corporate financing



Note: Newly issued equity and change in corporate loans (seasonally adjusted), corporate bonds.
Source: Bank of Thailand and Thai BMA

[1/] New issuance of private loans is the change in loan outstanding of all depository institutions (domestically-registered commercial banks, branches of foreign banks, international banking facilities, financial companies, specialized financial institutions, savings cooperatives, and money market mutual funds, excluding the Bank of Thailand). Therefore, transfers of loans between institutions are not included.

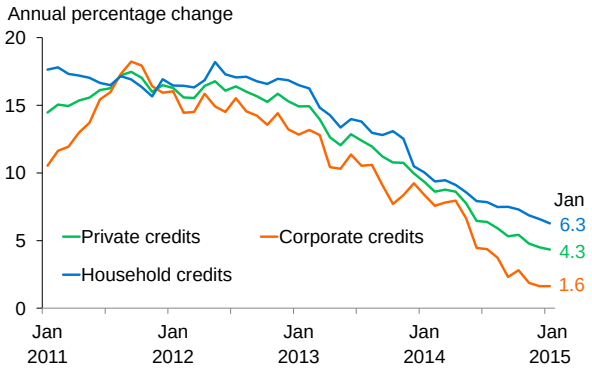
recovery and future income uncertainty. Lending standards for households, however, remained unchanged. Overall, private credits continued to moderate from the same period last year, from 5.4 percent at the end of October of 2014 to 4.3 percent at the end of January 2015 **(Chart 3.15)**.

Looking ahead, the Senior Loan Officers' Survey indicated that the demand for both business and household credits will pick up slightly. Car loans, however, will likely continue to decline. Lending standards are expected to remain unchanged from the previous period.

With regard to depository institutions, the number of special deposit products, as well as the deposit rates, declined moderately following sluggish demand for loans **(Chart 3.16)**. Deposit rates of the four largest commercial banks[2/] at the end of February 2015 were unchanged from the end of November 2014. Specifically, the 12-month deposit rates averaged at 1.73 percent.

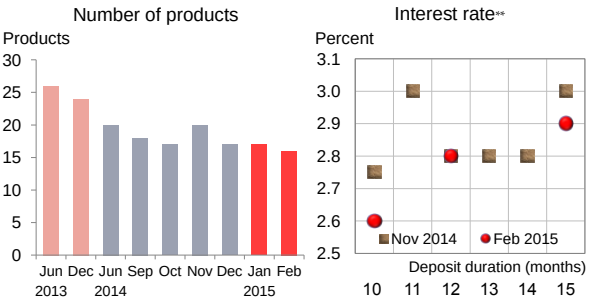
In 2014 Q4, new deposits[3/] grew, particularly those from household **(Chart 3.17**). The trend was a response to various types of special deposit products introduced in the previous quarter by financial institutions seeking additional funding amidst signs of economic recovery. The growth of deposits and bills of exchange (B/E) rose from 2.9 percent in October 2014 to 4.3 percent in January 2015 **(Chart 3.18)**. With deposit growth outpacing that of credits, commercial banks' loan to deposit

Chart 3.15 Other Depository Institutions' private credits



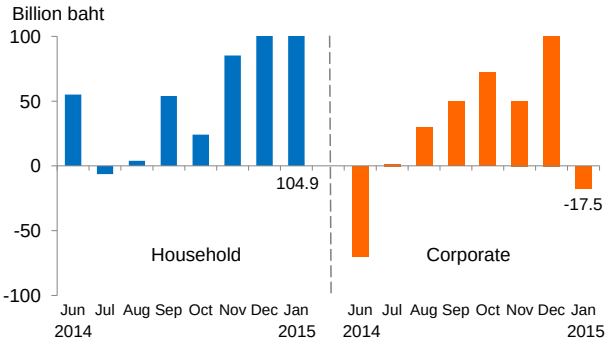
Source: Bank of Thailand

Chart 3.16 Special deposit products*



Note: *Offered by 8 commercial banks, i.e. Bangkok Bank, Krungthai Bank, KasikornBank, Siam Commercial Bank, Thai Military Bank, Tisco Bank, Kiatnakin Bank, as of 23 February 2015
**Highest interest rate from all products of the same duration offered by the 8 banks
Source: Bank of Thailand

Chart 3.17 New deposits*

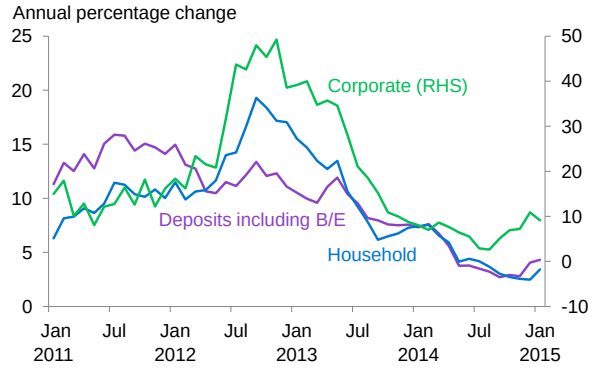


Source: Bank of Thailand

[2/] BBL, KTB, KBANK and SCB

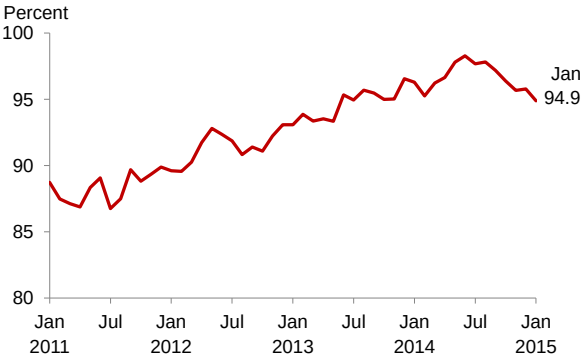
[3/] The amount of new deposits is calculated from the change in deposits at all depository institutions (excluding the Bank of Thailand). Therefore, it includes neither transfers of deposits among banks, nor rollovers of deposits.

Chart 3.18 Other Depository Institutions' deposits



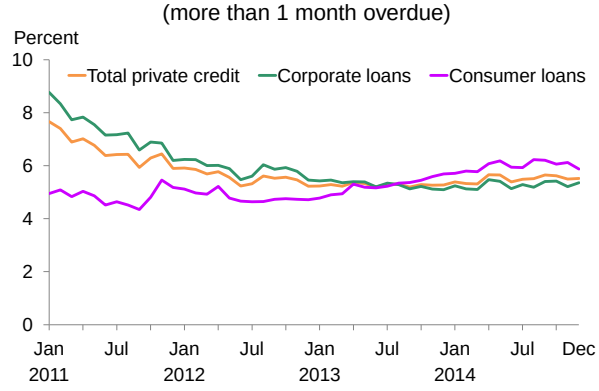
Source: Bank of Thailand

Chart 3.19 Loan to deposit and B/E ratio of commercial banks



Source: Bank of Thailand

Chart 3.20 NPL ratio of commercial banks (more than 1 month overdue)



Source: Bank of Thailand

including B/E ratio edged down to 94.9 at the end of January 2015 **(Chart 3.19)**.

Stability of financial institutions

Earnings of commercial banks and stability of financial institutions in 2014 Q4 remained robust, though the economic slowdown had somewhat weighed on credit growth. Credit quality remained unchanged, while commercial banks stepped up their loan loss provision and continued to maintain high levels of capital in order to cushion against economic uncertainty.

In 2014 Q4, commercial bank loans grew at 5.0 percent, down from 5.6 percent in the previous quarter. Overall loan quality remained unchanged, as reflected in the drop in the Delinquency and NPL Ratio to 5.5 percent, driven mainly by consumer credits **(Chart 3.20)**. Also, commercial banks' provisions remained high, indicating the commercial banks' ability to bear additional costs and deteriorated credit quality for loans. The ratio of actual loan loss provision to regulatory loan loss provision and the Capital Adequacy Ratio were recorded at 169.4 and 16.8 respectively.

3.3 Non-financial sectors

Stability of the non-financial sectors improved in line with the economic recovery in 2014 Q4. However, the economic recovery and its impact on the loan repayment ability of various sectors require ongoing monitoring since the recovery thus far has proved to be slower than expected due to depressed agricultural prices, elevated household debt, and subdued demands for real estate given future income uncertainty.

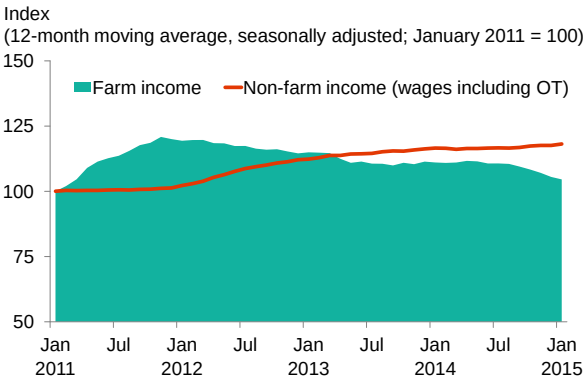
Household sector

Stability of the household sector remained solid, due to low risk of defaults. Nevertheless, repayment ability of low-income and agricultural households could be a source of concern, and thus warrants monitoring.

Stability of the household sector was well-maintained. Household debt slowed due to both consumers' careful spending behavior and banks' cautious lending standards. Yet, as the growth of household debt still outpaced GDP growth, household debt-to-GDP ratio increased moderately to 84.7 percent at the end of 2014 Q3 **(Table 3.2)**.

Overall, the fragility of the household sector remained contained and did not pose risks to the financial system. The NPL and delinquency ratio (share of loans overdue by over 1 month) of commercial banks declined from 6.2 percent at the end of 2014 Q3 to 5.9 percent at the end of 2014 Q4 **(Table 3.2)**. Nonetheless, loan repayment risks could rise among low-income and agricultural households **(Chart 3.21)**, given depressed agricultural and non-agricultural earnings.

Chart 3.21 Household income



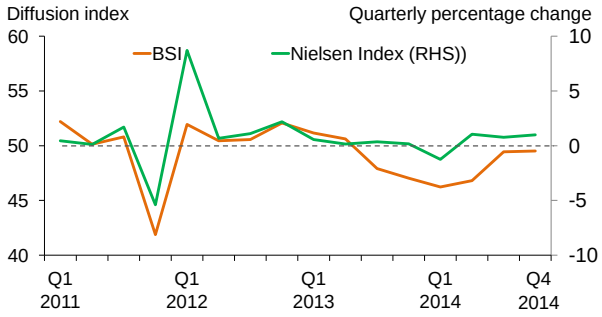
Source: Office of Agricultural Economics and National Statistical Office, calculations by the Bank of Thailand

In the coming periods, the MPC will closely monitor the recovery path of incomes and the repayment ability among these vulnerable groups.

Corporate sector

In 2014 Q4, stability of the corporate sector improved somewhat, owing partly to the overall economic recovery. For example, the retail sector expanded in line with a gradual recovery in private consumption. Despite improving retail sector, Business Sentiment Index stalled, hovering below 50 in the final quarter of 2014 **(Chart 3.22).** In addition, SMEs were faced with liquidity constraints and strict lending standards by commercial banks. As a result, they were more fragile compared to large corporate, and hence their loan repayment ability need to be monitored going forward.

Chart 3.22 Business Sentiment Index (BSI)* and Nielsen Index**



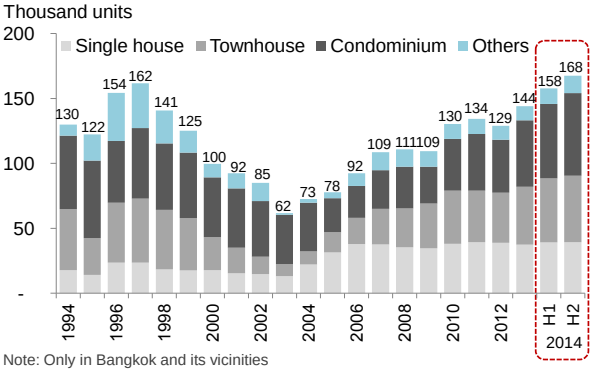
Note: *Diffusion Index (unchanged = 50)
**3-month moving agerage, seasonally adjusted
Source: Business Sentiment Survey taken in December 2014 by the Bank of Thailand

Meanwhile, the energy sector suffered from losses on oil stock and asset depreciation from the sudden plunge in global oil prices. However, the high ratio of retained earnings to total assets should help buffer the loss. Moreover, many firms have already taken measures to mitigate the oil price risk, for instance by implementing cost-cutting strategies and delaying investment plans.

Real estate sector

Real estate markets in Bangkok and its vicinities, especially for condominiums, may be running the risk of oversupply, due to slower-than-expected recovery in purchasing power. The issue thus requires close monitoring going forward.

Chart 3.23 Outstanding real estate supply at year end, classified by residential types, 1994-2014*



Note: Only in Bangkok and its vicinities
Source: Agency for Real Estate Affairs (AREA)

Demand for housing moderated towards the end of 2014 as a result of mounting concerns over future economic conditions and income

prospects. Persistently high household debt and caution on lending to the sector by financial institutions also contributed to the demand slowdown. Thus, the number of condominium ownership registrations contracted compared to the same period last year, while the cumulative excess supply of condominium in Bangkok and its vicinities continued rising **(Chart 3.23)**. As a result, income realization was delayed, hurting the liquidity of real estate firms. In the following period, real estate supply is expected to moderate, as some of the new project launches have be postponed **(Chart 3.24)**.

The quality of pre-finance and post-finance loans remained steady, suggested by largely unchanged Special Mention ratio of commercial banks. Also, the NPL Ratio edged down slightly as banks have already completed their loan restructuring **(Chart 3.25 and 3.26)**.

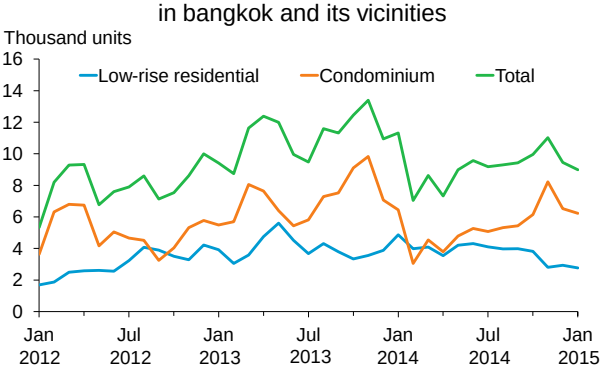
In the following period, the economic recovery and consumer confidence in future income prospects will be the key determinants of housing demand. It, therefore, remains to be seen whether the improvement in purchasing power will be sufficient to absorb existing excess supply.

Fiscal sector

The fiscal sector continued to be robust overall. The high level of treasury balance helped cushion the impact of lower-than-projected revenue collection. Meanwhile, the ratio of public debt to GDP decreased, after falling state enterprises' foreign debts as a result of the baht appreciation.

The fiscal sector remained healthy, as reflected by 1) the high level of treasury balance, which helped cushion the impact of lower-than-

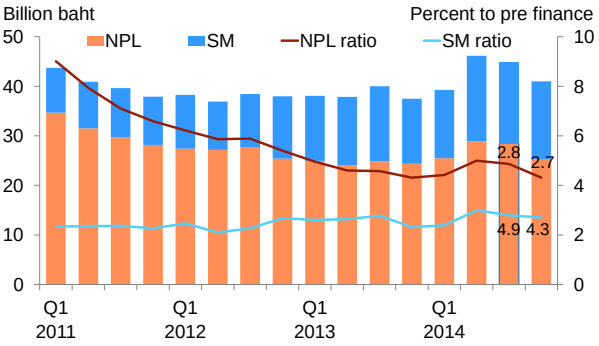
Chart 3.24 New residential project launches in bangkok and its vicinities



Note: *3-month moving average

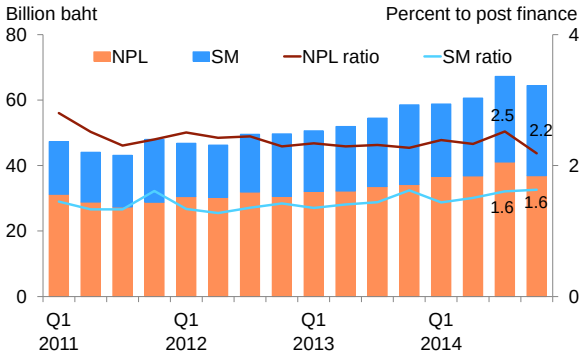
Source: Agency for Real Estate Affairs (AREA) and calculations by the Bank of Thailand

Chart 3.25 Mortgage loan quality of commercial banks (Pre finance)



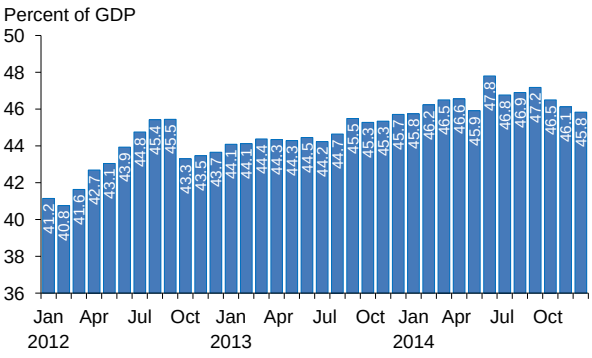
Source: Bank of Thailand

Chart 3.26 Mortgage loan quality of commercial banks (Post finance)



Source: Bank of Thailand

Chart 3.27 Public Debt to GDP ratio



Note: calendar years

Source: Public Debt Management Office

expected revenue collection. At the end of January 2015, treasury balance stood at 116 billion baht, a level sufficient to provide support for the government when needed; and 2) the ratio of public debt to GDP, which decreased from 47.2 percent in September to 45.8 percent in December 2014, well below the fiscal sustainability threshold of 60 percent **(Chart 3.27)**. The decline in the ratio was a result of three main factors, namely (1) debt repayment of financial and non-financial state-owned enterprises associated with the Fund for Rehabilitation of Foreign Debt Repayment of State-owned Enterprises, (2) bond-redemption by state-owned depository institutions, and (3) downward valuation change from currency appreciation in state enterprises' debt outstanding.

Looking ahead, a few risk factors need to be monitored. First, investment in Water Resource Management Project and Road Transportation Emergency Development Plan would raise the level of government debt. Second, the collection of personal income and corporate income taxes may be below target this year should the economic recovery continues to disappoint. In that case, the treasury balance will need to be tapped in order to fill the revenue gap, as was the case in 2014. The resulting decline in treasury balance will imply limited resouces for the government in the future. Nevertheless, slowdowns in investment from downward adjustments in auction prices for government projects should help ease pressure on the treasury balance in the short run.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2013	2014	2014				2015	
			Q1	Q2	Q3	Q4	Jan	Feb
1. Financial markets sector								
Bond market								
Bond spread (10 years-2 years)	1.0	1.3	1.5	1.4	1.2	0.9	0.5	n.a.
Equity market								
SET Index (End of period)	1,298.7	1,497.7	1,376.3	1,485.8	1,585.7	1,497.7	1,581.3	n.a.
Actual volatility (SET Index)[1/]	19.7	11.9	16.1	10.4	8.3	12.8	14.8	n.a.
Price to Earnings Ratio (times)	14.6	17.8	15.6	17.9	18.4	17.8	18.9	n.a.
FX market								
Actual volatility (baht) (%annualized)[2/]	6.2	3.9	4.6	3.8	3.8	3.5	3.5	n.a.
Nominal Effective Exchange Rate (NEER)	107.0	104.3	102.7	102.7	104.6	107.0	110.3	n.a.
Real Effective Exchange Rate (REER)	106.5	103.0	101.9	101.9	103.4	105.0	n.a.	n.a.
2. Financial institutions sector[3/]								
Minimum lending rate (MLR)[4/]	6.8	6.8	6.8	6.8	6.8	6.8	6.8	n.a.
12-month fixed deposit rate[4/]	2.2	1.7	1.7	1.7	1.7	1.7	1.7	n.a.
Capital adequacy								
Regulatory capital to risk-weighted asset (%)	15.7	16.8	15.5	15.9	17.1	16.8		
Earnings and profitability								
Net profit (billion baht)	203.9	214.2	51.8	59.8	53.8	48.7		
Return on assets (ROA)	1.33	1.32	1.29	1.48	1.33	1.18		
Liquidity								
Loan to deposit and B/E	96.6	95.7	95.9	97.9	97.2	95.7		
3. Household sector								
Household debt to GDP (times)	82.3	n.a.	82.8	83.5	84.7	n.a.		
Financial assets to debt (times)	2.0	n.a.	2.0	2.0	2.0	n.a.		
NPL and delinquency ratio (%)								
Thai commercial banks :								
Consumer loans	5.7	5.9	5.8	5.9	6.2	5.9		
Mortgage loans	3.9	3.8	3.8	3.8	4.1	3.8		
Auto leasing	9.4	10.8	9.9	10.5	10.6	10.8		
Credit cards	4.7	5.3	5.3	5.5	6.1	5.3		
Other personal loans	4.6	5.0	4.8	5.0	5.6	5.0		
4. Non-financial corporate sector [5/]								
Operating profit margin (%)	5.8	n.a.	4.7	4.5	4.7	n.a.		
Debt to equity ratio (times)	0.8	n.a.	0.8	0.8	0.8	n.a.		
Income coverage ratio (times)	6.2	n.a.	6.1	5.8	5.7	n.a.		
Current ratio (times)	1.5	n.a.	1.6	1.5	1.5	n.a.		

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2013	2014	2014				2015	
			Q1	Q2	Q3	Q4	Jan	Feb
5. Real estate sector								
The number of approved mortgages from banks (Bangkok and its vicinity)	71,701	62,839	12,880	16,315	17,345	16,299	2,688	n.a.
Single-detached and semi-detached houses	18,353	15,694	3,331	4,206	4,230	3,927	657	n.a.
Townhouses and commercial buildings	25,261	21,764	4,784	5,921	5,844	5,215	1,058	n.a.
Condominiums	28,087	25,381	4,765	6,188	7,271	7,157	973	n.a.
The number of new openings (Bangkok and its vicinity)	131,550	111,211	25,862	28,714	28,268	28,367	6,785	n.a.
Single-detached and semi-detached houses	17,226	18,933	4,754	5,657	5,349	3,173	863	n.a.
Townhouses and commercial buildings	30,074	26,980	7,499	7,261	6,611	5,609	842	n.a.
Condominiums	84,250	65,298	13,609	15,796	16,308	19,585	5,080	n.a.
Housing price index[6/]								
Single-detached houses (including land)	111.5	117.0	114.7	115.5	119.3	118.4	118.1	n.a.
Townhouses (including land)	121.3	130.2	125.7	128.6	134.2	132.5	131.9	n.a.
Condominiums	141.7	154.3	146.2	153.1	156.0	161.9	164.1	n.a.
Land	138.3	150.7	145.9	148.3	153.0	155.5	159.0	n.a.
6. Fiscal sector								
Public debt to GDP (%)	45.7	45.8	46.5	47.1	47.2	45.8	n.a.	n.a.

[1/] Daily volatility (using exponentially weighted moving average method)
[2/] Annualized standard deviation of return
[3/] Based on data of all commercial banks
[4/] Average value of 4 largest Thai commercial banks
[5/] Only listed companies on SET (median)
[6/] Based on data of new approvals by commercial banks using hedonic regression method (January 2009 = 100)
(Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change)

Real interest rate: concept and calculation

In the current environment of falling inflation in line with the global lower crude oil prices, real interest rate is increasingly being used in the analysis of monetary condition and financing costs. Nonetheless, different calculation methods of real interest rate have resulted in differing views on monetary condition and some misinterpretation of monetary policy direction. This article aims to establish an accurate understanding of real interest rates, and this in turn should help enhance the effectiveness of the BOT's communication concerning the real interest rate and monetary policy stance, consistent with the monetary policy objectives.

Real interest rate

Real interest rate is the nominal interest rate minus the expected inflation. It is an important factor in the decision-making process of economic agents such as firms, households and other sectors because it better reflects the true opportunity cost of present consumption relative to future consumption. In other words, real interest rate represents future interest income in the form that can be used to purchase real goods and services (real purchasing power).

In allocating their budget between spending and saving, most households form expectations on future prices of goods in order to ensure that future income is adequate to meet future expenses and expected inflation. Similarly, firms make their borrowing and investment decisions based on their expectation of future prices. For example, retail businesses may decide to borrow at the current nominal rate to finance their business expansion, in anticipation of higher revenue as the economy expands. **It is clear that the expectation of *future* prices of goods influence household consumption and business investment decisions in the *current* period. Therefore, it is the expected inflation, rather than current level of inflation, which should be used in the calculation of real interest rates.**

How is expected inflation assessed?

Expected inflation is typically derived via four main methods: (1) household and business surveys on the outlook of future prices of goods and services; (2) surveys of professional forecasters working in the financial markets; (3) implied inflation calculated from inflation-linked bonds; and (4) central bank forecasts.

Preliminary studies show that most central banks conduct household and business surveys to assess the outlook of prices of goods and services.[1/] However, in their communication with the public, several central banks use expected inflation obtained from a variety of methods. They also regularly refer to the inflation expectations via various channels of communication, including monetary policy reports and speeches. Some central banks not only communicate the developments of expected inflation, but also discuss the limitations of the various methods used to assess expected inflation. For example, at its meeting on January 27-28, 2015,[2/] the Federal Open Market Committee (FOMC) deliberated at length on the expected inflation issues because different calculation methods produced differing results. In the view of some FOMC members, although surveys suggested a rather stable outlook of expected inflation, the implied inflation

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[1/] Central banks that conduct such surveys include those in the U.S., England, the euro area, South Korea and Indonesia.

[2/] From the minutes of FOMC meeting on January 27-28, 2015 (available on the website: http://www.federalreserve.gov/monetarypolicy/fomccalendars.htm)

(breakeven inflation) derived from the Treasury Inflation Protected Securities (TIPs)[3/] pointed to lower expected inflation. They went on to argue that the stability of the survey-based measures should not be taken as providing much assurance, as Japan in fact already experienced deflation in the late 1990s and early 2000s even as their survey-based measures of long-term inflation expectation had not indicated sustained deflation. Moreover, it remained unclear whether the decline in breakeven inflation derived from the TIPs yields was being reflected solely by lower expected inflation. Therefore, such information should be used with caution in monetary policy deliberations.

Calculations of Thailand's expected inflation

Chart 1 Thailand's real interest rate



Source: Nominal policy interest rate as available on the BOT website, expected inflation from Consensus Economics Inc. survey of professional forecasters

The real policy interest rate is the nominal policy interest rate minus the expected inflation. Since 2009, Thailand's nominal policy interest rate has remained below expected inflation, resulting in a period of sustained negative real interest rates **(Chart 1)**.

As part of its monetary policy consideration, the MPC monitors the development of real interest rates obtained by a variety of methods in obtaining the expected inflation, recognizing the limitations of each method. For example, household, business and forecasters surveys may not have enough frequency and some respondents may attach far too considerable weight to the current level of inflation. Also, the expected inflation derived from the inflation-linked bonds may be somewhat volatile if the market is illiquid. At present, the real interest rates obtained through surveys of professional forecasters and business surveys are comparable and move in line with each other.

The MPC not only pays attention to expected inflation, but also monitors the influence of current prices of goods and services on corporate and household sectors. The larger-than-anticipated fall in the prices of goods has caused some firms' revenues to drop sharply than previously expected. However, interest expenses have not declined accordingly, and therefore potentially undermined firms' ability to repay debt in the current period and lead to tighter financial conditions. Therefore, both current prices of goods and services and expected inflation can influence income, interest expenses and financial conditions of households and firms.

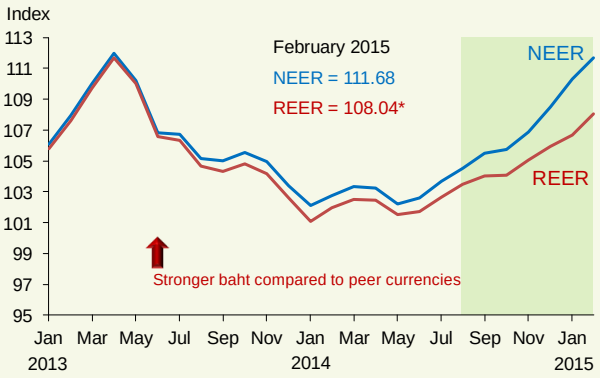
In deliberating monetary policy, the MPC takes into account the issues above and closely monitors information on changing financial conditions. The MPC communicates with the public through various channels, including the MPC edited minutes, the monthly Economic and Monetary Conditions and quarterly Monetary Policy Reports, to ensure efficient and effective monetary policy communication.

[3/] Calculations made in a model in order to separate the different factors that make up the yields of inflation-linked bonds.

Strong baht: causes, impacts, and adaptations by Thai businesses

Amidst the recent Thai baht appreciation trend, concerns are mounting over its potential impact on exports and economic recovery. Since the end of 2014, the continuous appreciation of the Nominal Effective Exchange Rate (NEER) as well as the Real Effective Exchange Rate (REER) implied stronger baht relative to major and regional currencies[1/] **(Chart 1).** Such development was due in large part to a sharp depreciation of the currencies of Thailand's trading partners and competitors as a result of country-specific factors which in turn reflected the fragility of their respective economies in the following aspects;

Chart 1 Nominal Effect Exchange Rate and Real Effective Exchange Rate



Note: *Preliminary data computed from February 2015 NEER and January 2015 CPI, as proxy for February 2015 CPI.
Source: Bank of Thailand

1.1 The euro and the yen depreciated due to continued monetary policy easings, specifically a larger-than-expected quantitative easing (QE) program by the European Central Bank (ECB), a series of QE measures by the Bank of Japan, and the lifting of the Swiss franc-euro exchange rate cap.

1.2 Regional currencies weakened following the fall in oil prices as well as commodity prices. The prospect of a worsening export revenues and fiscal position of oil-exporting countries has led to a depreciation of the Indonesian rupiah, the Malaysian ringgit and the Russian ruble. Similarly, the currencies of other commodity exporters, such as the Australian dollar, lost value on the back of falling commodity prices.

Following a plunge in global oil prices, Thailand, as a net oil-importing country, saw an improvement in its current account as its oil imports accounted for a relatively large share of the GDP compared to its regional peers.[2/] As a result, Thailand experienced a sharp fall in import values, while export values remained largely unchanged from the previous period[3/],. Moreover, foreign investor confidence in sound fundamentals of the Thai economy helped attract foreign direct investment. With additional QE by the ECB and monetary easing policies in other economies, more capital flows are expected to move into Thai assets going forward.

The Monetary Policy Committee, however, judged that the impact of the current round of ECB's QE should not be as pronounced as those of the Fed for the following reasons.

(1) Capital flows from the euro area into regional markets have not increased significantly in recent periods, despite the implementation of QE by the ECB.

[1/] NEER and REER rose by 6.8 and 4.4 percent, respectively, from August 2014 to February 2015.

[2/] Average Net Oil Imports to GDP ratio between 2011-2013 for South Korea, Thailand, Indonesia, and Malaysia are 9, 8, -2.3, and -5.5 percent, respectively.

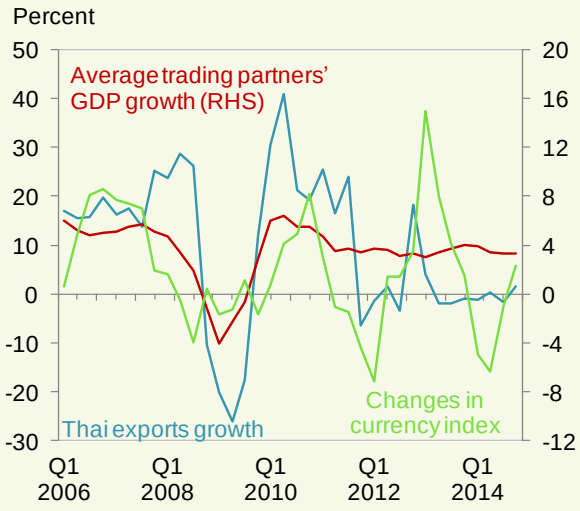
[3/] Total export values in 2013 and 2014 declined by 0.2 and 0.3 percent from the previous year.

(2) The transmission of liquidity from European to Asian banks is limited, because banks in the euro area are too fragile to take on additional risks by lending to Asian banks despite their improved liquidity positions.

(3) Portfolio investors in the euro area are known to prefer to invest in countries with low risks and high credit ratings, rather than in emerging markets.

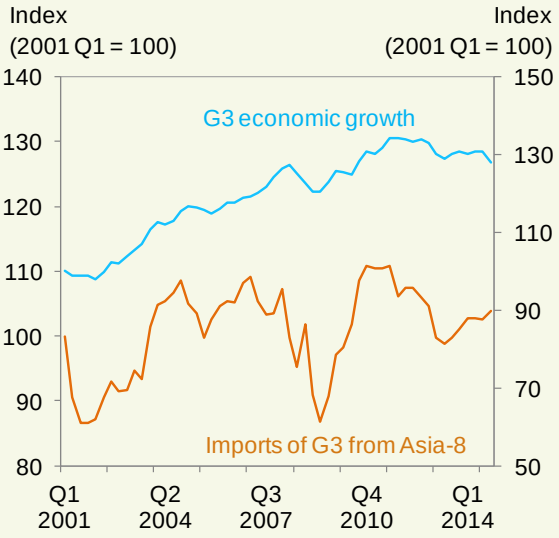
The baht appreciation trend has triggered concerns among some parties over its adverse impact on the already weak exports. An analysis of historical data reveals that the key determinant of Thai exports is the economies of our trading partners[4/] **(Chart 2).** Unfortunately, the ongoing slowdown of our major trading partners' economies, including the U.S., euro area, Japan, and China have stymied demand for exports from Asia, leaving regional economies to increasing rely on their domestic demand[5/] for growth. The declining exports also resulted from a number of other external and internal factors. First, under the changing global trade landscape, Asian exporters stand to benefit less from recovery of G3 countries **(Chart 3).** Second, shifting consumer preferences have rendered certain products obsolete. Thus, producers of those products have experienced slower export growth. This is especially the case for high-technology sectors. As for Thailand, the chronic lack of investment in new technology has led to structural constraints that prevent Thai firms from producing products in demand, such as smart phones and tablets.

Chart 2 Thai exports growth, trading partners' GDP growth, and currency index changes



Source: Bank of Thailand

Chart 3 Relationship between G3 economic growth and imports from Asia-8



Note: Asia-8 includes Hong Kong, Taiwan, South Korea, Singapore, Malaysia, Indonesia, Philippines, and Thailand

Source: CEIC

[4/] The correlation between export values and trading partners' growth is 0.75, whereas that between export values and changes in currency index is 0.26. The calculation was based on quarterly data from 2006 to 2014.

[5/] The correlation between G3 economic growth and the group's import from Asia-8 edged down from 0.88 (for 2001 Q1 to 2007 Q4) to 0.51 (for 2010 Q1 to 2014 Q4).

Although the value of Thai baht is not the main determinant of export volume, its rapid appreciation may hurt certain industries, particularly SMEs with fragile financial conditions and liquidity constraints during the economic slowdown. Firms that face intense price competition and low profit margins are also likely to be affected. The impact of a currency appreciation on Thai exports can be summarized by 2 main channels as follows **(Chart 4).**

Chart 4 Impacts of foreign exchange fluctuations and managements of foreign exchange risks

Industries	Impacts of exchange rate fluctuations			Management of exchange rate risks			
	Lower converted revenues	Higher costs in baht	Loss of price-competitiveness	FX hedging instruments	Price negotiations	Adjustments in payment schedule	Cost-revenue currency matching
Agricultural	/		/	/			
Food	/			/	/		
Textiles	/		/				
Furniture	/				/		
Electronics					/		/
Electrical appliances							/
Automotive				/	/	/	
Metals		/			/		
Chemical products		/					

Source: Interviews of various industry associations

1. As export revenues are denominated in foreign currencies, the baht appreciation will lead to lower revenues in baht term. This hurts firms' liquidity and financial positions, most significantly in sectors with high currency mismatch between revenues and costs. These are usually sectors that utilize domestic raw materials, rely heavily on export revenues, but are not fully hedged against currency risks, such as agricultural, textile, and furniture industries. To make matters worse, firms in these sectors are often small with limited bargaining powers. Therefore, they are unable to adjust prices in order to mitigate the impact of currency appreciation.

2. Ability of Thai exporters to engage in price competition has deteriorated. Export volumes in highly price-competitive sectors, such as agricultural and textile, are declining as a result. Because products in these sectors tend to be substitutable, lower prices are key to export success. If the currency of a rival country depreciates, in turn making its goods cheaper, Thai exporters may immediately lose their market shares. Fortunately, however, the share of exports of these price-sensitive sectors is gradually declining as domestic labor costs rose.

Aside from sensitive industries mentioned above, the effect of strong baht will be limited in most industries, especially those in global production chains, such as automotive and parts and electronics parts. Due to the nature of the contracts and the relationship between parent companies and Thai firms within the production chains, it is difficult for the parent companies to switch to partners in a different country. In addition, during each trading round, Thai producers can negotiate for the parent companies to shoulder some of the currency impacts. Furthermore, a strong baht helps reduce production costs for industries with high import contents, such as metals and chemical products.

Although firms in some industries will likely be affected by a strong baht, the overall impact is considered much smaller than in the past, due to various forms of adaptations. The following are three important adaptations which helped Thai companies mitigate the impacts of currency fluctuations.

1. Prudent management of foreign-currency revenues

Thai firms employ various strategies in order to mitigate the impact of strong baht. For example, they can match the currencies of their revenues to those of their costs, thus creating a so-called natural hedge. Those in electrical appliances, machineries, and automotive industries can adjust payment schedules according to currency movements at the time. Moreover, many firms continue to seek new markets for exports, while others choose to relocate production bases in foreign countries in order to benefit from lower wages.

2. Use of exchange rate hedging instruments

Although FX hedging instruments, such as forwards or options, come with costs, they can effectively reduce risks associated with foreign exchange fluctuations. Currently, firms that employ these instruments are mostly multinationals or large Thai corporates. Meanwhile, only a small, though increasing, fraction of small and medium firms use these products regularly due to lack of understanding and access. Realizing such limitations, the Bank of Thailand, together with commercial banks, has strived to encourage greater use of hedging instruments by providing more information and developing affordable products that would best fit the demand of business owners.

3. Development in manufacturing capacity to produce high-quality and distinct products.

This final measure is crucial in fostering resiliency to currency fluctuations in a sustainable manner. Typically, firms whose products compete mainly on the basis of quality will enjoy high bargaining powers. They are therefore guaranteed favorable prices without having to rely on baht weakness. South Korea provides an example of successful development of its exports sector via this avenue. Last but not least, the current period of relatively strong baht provides an opportunity for Thai firms to import capital goods in order to upgrade their production capacities. Through such productivity enhancement, the Thai firms will eventually overcome the structural problems that have been plaguing Thai exports.